

Regeneration following tissue necrosis is mediated by non-apoptotic caspase activity

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eLife Assessment

This manuscript reports **fundamental** discoveries on how necrotic cells contribute to organ regeneration through apoptotic signalling to produce cells with non-lethal apoptotic caspase activity that contribute to the regenerated tissue. These findings will be of broad interest to those who study wound repair and tissue regeneration. The strength of the evidence is **solid** and has been improved in the revised version.

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Abstract

Tissue necrosis is a devastating complication for many human diseases and injuries. Unfortunately, our understanding of necrosis and how it impacts surrounding healthy tissue – an essential consideration when developing effective methods to treat such injuries – has been limited by a lack of robust genetically tractable models. Our lab previously established a method to study necrosis-induced regeneration in the *Drosophila* wing imaginal disc, which revealed a unique phenomenon whereby cells at a distance from the injury upregulate caspase activity in a process called Necrosis-induced Apoptosis (NiA) that is vital for regeneration. Here we have further investigated this phenomenon, showing that NiA is predominantly associated with the highly regenerative pouch region of the disc, shaped by genetic factors present in the presumptive hinge. Furthermore, we find that a proportion of NiA fail to undergo apoptosis, instead surviving effector caspase activation to persist within the tissue and stimulate reparative proliferation late in regeneration. This proliferation relies on the initiator caspase Dronc, and occurs independent of JNK, ROS or mitogens associated with the previously characterized Apoptosis-induced Proliferation (AiP) mechanism. These data reveal a new means by which non-apoptotic Dronc signaling promotes regenerative proliferation in response to necrotic damage.

Introduction

Necrosis is the rapid, disordered death of cells characterized by the loss of membrane integrity and release of cytoplasmic contents into the surrounding tissue (Hajibabaie *et al.* 2023). This catastrophic type of cell death can occur in diverse tissues and is central to many human conditions, particularly those related to ischemic injuries. Such conditions include chronic illnesses like diabetes, joint disorders, sickle-cell anemia and other inherited and congenital diseases (Masi *et al.* 2007, Mulay *et al.* 2016, Karsch-Bluman *et al.* 2019, Tonnus *et al.* 2021, Li *et al.* 2023), as well as more acute medical events like strokes, heart attacks, bacterial infections and common traumatic injuries (Konstantinidis *et al.* 2012, Hakkarainen *et al.* 2014, Bonne and Kadri 2017, Wu *et al.* 2018). Even therapeutic interventions, in particular treatments for cancer, can result in this devastating form of damage (Robertson *et al.* 2017, Nakada *et al.* 2019, Yang *et al.* 2021). Unfortunately, current strategies to treat necrosis mainly focus on invasive procedures that are often met with limited success. With such substantial bearings on human health, it is crucial to better understand the effects of necrosis in disease and injury, particularly in the context of tissue repair and regeneration.

We currently have a limited understanding of how necrosis impacts surrounding healthy tissue during wound healing. Indeed, much of our understanding about how cell death influences tissue repair comes instead from models involving programmed cell death (PCD) like apoptosis (Hajibabaie *et al.* 2023). This highly regulated process can be triggered by intrinsic or extrinsic pathways, both of which lead to the activation of caspases that mediate the controlled disassembly of the cell (Ashkenazi and Salvesen 2014). Studies of PCD in a variety of species have shown that cells undergoing apoptosis can release signaling molecules that are interpreted by surrounding tissues to drive wound healing events, such as tissue remodeling, immune responses, survival and proliferation of surrounding cells (Tseng *et al.* 2007, Fan and Bergmann 2008, Chera *et al.* 2009, Bergmann and Steller 2010, Li *et al.* 2010, Pellettieri *et al.* 2010, Ryoo and Bergmann 2012, Vriza *et al.* 2014, Fuchs and Steller 2015, Perez-Garijo and Steller 2015, Fogarty and Bergmann 2017, Perez-Garijo 2018). For example, the signaling molecules Prostaglandin E2 and Hedgehog are produced by dying hepatocytes to induce regenerative proliferation in the vertebrate liver (Jung *et al.* 2010, Li *et al.* 2010), while apoptotic-deficient mice show both impaired liver regeneration and epidermal recovery after wounding (Li *et al.* 2010). Although mitogenic signaling by apoptotic cells is an established and conserved process, whether similar signaling events occur following necrotic cell death is less clear.

Evidence of apoptotic signaling first originated from studies of the larval wing primordia in *Drosophila* (Perez-Garijo *et al.* 2004, Ryoo *et al.* 2004). This epithelial tissue has been extensively characterized as a model for growth, development and regeneration, including the role that cell death plays in these processes (Beira and Paro 2016, Worley and Hariharan 2022). Ongoing studies of this model have identified an essential signaling network centered on the highly conserved JNK pathway. JNK activates several major signaling pathways including Hippo and JAK/STAT, which have conserved roles in promoting regeneration across species (Worley *et al.* 2012, Hariharan and Serras 2017, Fox *et al.* 2020, Worley and Hariharan 2022), as well as activating JNK itself via overlapping positive feedback loops. One such feedback loop acts through the initiator caspase Dronc (*Drosophila* Caspase-9), which, independent of its role in apoptosis, is translocated to the cell membrane to activate the release of ROS from the NADPH oxidase Duox (Amcheslavsky *et al.* 2018). ROS attracts hemocytes to further activate JNK signaling in the disc through the release of the TNF ligand Eiger (Fogarty *et al.* 2016, Diwanji and Bergmann 2018). In a related pathway, JNK can also lead to the expression of the Duox maturation factor *moladietz* (*mol*), thus activating this feedback loop without Dronc (Khan *et al.* 2017, Pinal *et al.* 2018). An important advance in elucidating this network was the ability to generate “undead cells”, using the baculovirus caspase inhibitor P35 to prevent apoptotic cells from dying (Hay *et al.* 1994). These

cells therefore persist, releasing mitogenic factors including Wingless (Wg, Wnt1), Decapentaplegic (Dpp, BMP2/4), Spitz (Spi, EGF) or Hedgehog (Hh) (Huh *et al.* 2004 [↗](#), Perez-Garijo *et al.* 2004 [↗](#), Ryoo *et al.* 2004 [↗](#), Perez-Garijo *et al.* 2005 [↗](#), Fan and Bergmann 2008 [↗](#), Perez-Garijo *et al.* 2009 [↗](#), Morata *et al.* 2011 [↗](#), Fan *et al.* 2014 [↗](#)). These signals subsequently promote proliferation of the surrounding cells in a phenomenon known as Apoptosis-induced Proliferation (AiP) (Fan and Bergmann 2008 [↗](#), Ryoo and Bergmann 2012 [↗](#), Fogarty and Bergmann 2017 [↗](#)).

By contrast, the genetic events following necrosis are less well explored. Necrosis is characterized by swelling and loss of cellular membrane integrity, with the release of cellular contents into the intercellular space causing a significant inflammatory response (Festjens *et al.* 2006 [↗](#), D'Arcy 2019 [↗](#), Hajibabaei *et al.* 2023 [↗](#)). Necrosis is highly variable, occurring as a regulated process, for example necroptosis, or as unregulated, caspase-independent lysis (Ashkenazi and Salvesen 2014 [↗](#), D'Arcy 2019 [↗](#)). The factors released from necrotic cells are collectively termed Damage-Associated Molecular Patterns (DAMPs) (Venereau *et al.* 2015 [↗](#), Roh and Sohn 2018 [↗](#)), which are thought to interact with pattern recognition receptors (PRRs) on nearby cells, mostly of the Toll-like receptor (TLR) family (Ming *et al.* 2014 [↗](#), Gong *et al.* 2020 [↗](#)). DAMPs are understood to mainly consist of fundamental cellular components like histones, chromatin, and actin (Venereau *et al.* 2015 [↗](#), Gordon *et al.* 2018 [↗](#), Roh and Sohn 2018 [↗](#)), although specific factors have also been described. For example, High-mobility group box 1 (HMBG1) has been characterized as a DAMP in models of spinal cord, cardiac and muscle injury where it promotes angiogenesis, attracts repair cells and induces proliferation (Venereau *et al.* 2015 [↗](#)), as well as in *Drosophila* models of necrosis (Nishida *et al.* 2024 [↗](#)). However, the overall role of DAMPs and how they influence healing and regeneration through interaction with healthy tissues has yet to be fully explored.

To better investigate necrosis-induced wound repair and regeneration, our lab developed a method to rapidly and reproducibly induce necrotic cell death within the developing *Drosophila* wing imaginal discs (Klemm *et al.* 2021 [↗](#)). Using a genetic ablation system we previously established, named Duration and Location (DUAL) Control (Harris 2023 [↗](#)), we can induce necrosis in the wing disc via expression of a leaky cation channel *GluR1^{LC}* (Liu *et al.* 2013 [↗](#), Yang *et al.* 2013 [↗](#)). Using this system (*DC^{GluR1}*), we showed that wing discs are capable of fully regenerating following necrotic injury at a rate comparable to that of damage induced by apoptosis (Klemm *et al.* 2021 [↗](#)). However, while apoptotic ablation leads to JNK signaling and extensive caspase activity, we found that necrosis leads to only minor levels of JNK-mediated apoptosis, which is confined to the wound edge, but unexpectedly generates significant caspase activity in cells distant from the injury. We called this non-autonomous caspase activation Necrosis-induced Apoptosis (NiA) (Klemm *et al.* 2021 [↗](#)). Unlike normal apoptotic cells, NiA form entirely independent of JNK signaling, and cannot be made undead using P35. We also demonstrated that NiA is essential for regeneration, although the mechanism was unclear.

Here we have further characterized the NiA phenomenon, finding that only regeneration-competent areas of the wing disc can produce NiA following damage, in part due to WNT and JAK/STAT signaling in the hinge that limits NiA to the pouch. Building upon our finding that NiA is necessary for regeneration, we show that NiA leads to localized proliferation significantly later in regeneration than previously observed. Using tools to trace caspase activity and cell death, we demonstrate that this is possible because a proportion of NiA survive effector caspase activation and persist late into regeneration where they promote proliferation.

Finally, we show that this proliferation relies on the initiator caspase Dronc, but surprisingly does not involve established AiP mechanisms. Our data suggest a model in which necrotic injuries induce caspase activity in cells at a distance from the injury, some of which undergo JNK-independent apoptosis (NiA), while others survive and promote proliferation through a novel non-apoptotic function of Dronc, which is separate from its role in AiP. We refer to these surviving NiA

cells as Necrosis-induced Caspase-Positive (NiCP) cells. These findings reveal an important genetic response to lytic cell death that could potentially be leveraged to augment regeneration of necrotic wounds.

Results

Formation of NiA occurs primarily in the wing pouch

Previously, we found that NiA occurs in the lateral pouch (LP) upon induction of necrosis in the distal pouch with *DC^{GluR1}* (Figure 1A, D and E, yellow arrowheads in E) (Klemm *et al.* 2021). The wing disc itself comprises different identities reflecting the adult structures they ultimately create, including the pouch, hinge and notum, which are themselves divided into compartments; anterior/posterior and dorsal/ventral (Figure 1B). Since these various disc identities have distinct regenerative capacities stemming from their different genetic responses to damage (Martin *et al.* 2017), to better understand the formation of NiA and the role it plays in regeneration we tested whether necrosis occurring in different areas of the disc leads to NiA. To do so, we utilized *GAL4/UAS/GAL80^{ts}* to conditionally express *UAS-GluR1^{LC}* (Liu *et al.* 2013, Yang *et al.* 2013) in the pouch, hinge or notum tissues (Figure 1C). As an initial test, we attempted to recapitulate our original observations made using *DC^{GluR1}* by employing an enhancer of the *spalt* gene driving *GAL4 (R85E08^{ts}>GluR1)* to cause necrosis in the distal pouch (Figure 1F and G). As anticipated, NiA are formed in the lateral pouch following 20 hr of ablation (denoted as 0 hr, when larvae are downshifted to 18°C) (Figure 1G). In this figure and others, NiA are recognized as cells positive for the cleaved caspase cDcp-1 and negative for GFP that labels the ablation domain (*UAS-GFP*) (Figure 1E and G, yellow arrowheads), which indicates that these caspase-positive cells originate outside the area of ablation. This test confirms that the NiA phenomenon occurs independent of the ablation system used. Notably, NiA are consistently absent from the presumptive hinge region surrounding the pouch following both *R85E08^{ts}>GluR1* or *DC^{GluR1}* ablation (Figure 1E and G). Indeed, outside of the pouch, cDcp-1 staining is only observed in a small area of the posterior pleura (Figure 1E, red arrowhead) and at low levels stochastically across the disc due to stress induced by temperature changes (Klemm *et al.* 2021). To further investigate the extent to which necrotic pouch tissue can induce NiA we next ablated the entire pouch using *rotund-GAL4 (rn^{ts}>GluR1)* or *nubbin-GAL4 (nub^{ts}>GluR1)* (Figure 1H, I and J, figure 1-figure supplement 1A). Necrosis of the whole pouch with either driver fails to result in NiA (Figure 1I and Figure 1-figure supplement 1A), even after 24 hr of recovery (Figure 1J). Importantly, the majority of cDcp-1-positive cells overlap with expression of the JNK target *Mmp1* (Figure 1I and J), and therefore resembles cells undergoing JNK-mediated apoptosis, such as those seen at the wound edge (WE) following distal pouch ablation (Figure 1 – figure supplement 1B, arrowhead) (Klemm *et al.* 2021). These data suggest that cells outside of the pouch are generally unable to respond to DAMPs released by pouch cells undergoing necrosis to generate NiA, or that such DAMPs are spatially limited.

To test whether necrosis in areas outside of the pouch can induce NiA, we ablated the proximal notum using *pannier-GAL4 (pnr^{ts}>GluR1)* (Figure 1K), using the absence of the notum Wg stripe to confirm loss of this area (Figure 1L and M, open arrowheads). In this ablation we observed only minimal NiA, with sporadic cDcp-1-positive, GFP-negative cells in the unablated areas of the notum (Figure 1L), which does not change after 24 hr (Figure 1M). To test the hinge, we used a putative *zfh1* enhancer driving *GAL4 (R73G07^{ts}>GluR1)*, which has hinge-specific expression (Figure 1N). Ablation of the hinge fails to generate NiA in the notum, and surprisingly does not induce a response in the neighboring pouch cells (Figure 1O and P), despite their demonstrated ability to form NiA (Figure 1G). Given that the regenerative capacity of the wing disc is mainly constrained to the pouch (Martin *et al.* 2017, Martin and Morata 2018), it is possible that the intrinsic differences in regenerative capacity across the disc accounts for the disparity in NiA/NiCP occurrence in the pouch versus the notum. A difference in the availability of

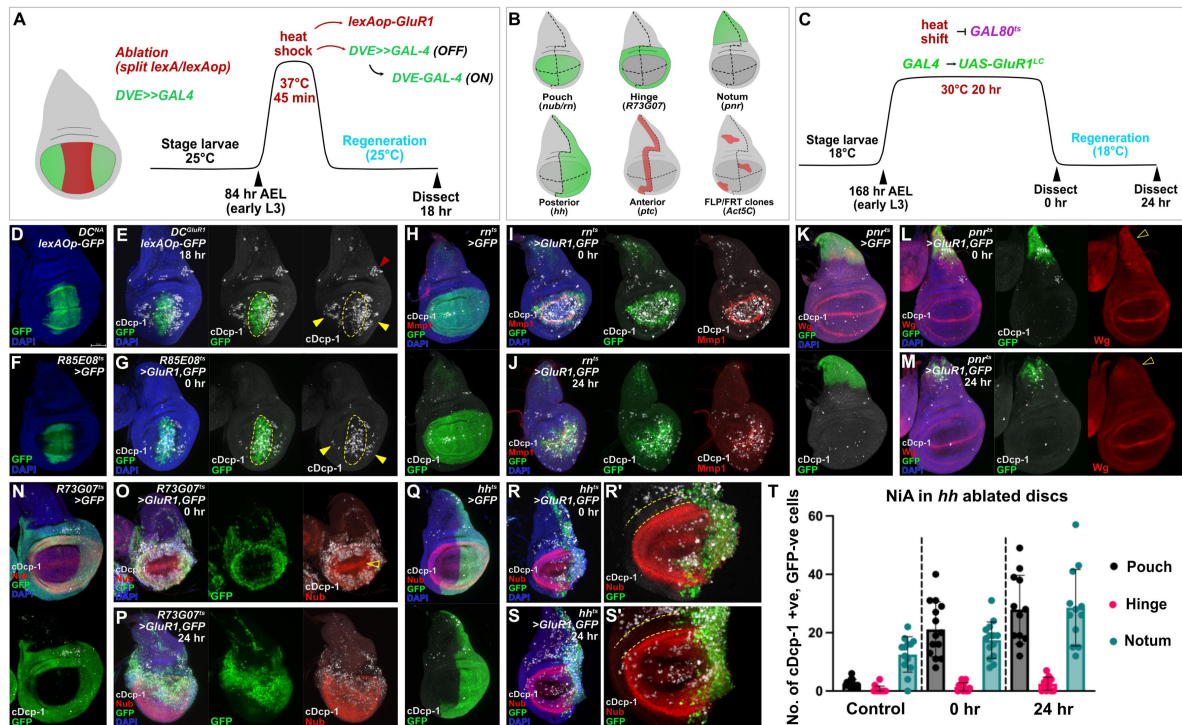


Figure 1.

The NiA response is restricted to pouch cells following ablation.

(A) A schematic of the *DC^{GluR1}* ablation scheme. This system utilizes a split LexA transcriptional activator that promotes the expression of *lexAop-GluR1* following a short heat shock (45 min. at 37°C). (B) A schematic of the different tissue domains and compartments targeted for *GAL4/UAS/GAL80^{ts}* (*GAL4^{ts}*) ablation. (C) A schematic of the temperature scheme used to induce necrosis with *GAL4^{ts}>GluR1* following the heat shift (20 hr at 30°C) induced inactivation of *GAL80^{ts}*. (D) A control disc for DUAL Control (*DC^{NA}*) bearing *lexAop-GFP*, which labels the domain targeted for ablation in *DC^{GluR1}* experiments. (E) A *DC^{GluR1}* ablated disc at 18 hr of regeneration bearing *lexAop-GFP*, yellow arrowheads indicate NiA cells, red arrowhead indicates potential NiA response in the posterior pleura. (F) A control disc for *R85E08-GAL4^{ts}* (*R85^{ts}>GFP*) ablation. (G) An *R85E08^{ts}>GluR1,GFP* ablated disc, arrowheads indicate the presence of NiA cells following ablation. (H) A control disc for whole pouch (*rn^{ts}>GFP*) ablation. (I-J) A *rn^{ts}>GluR1,GFP* ablated disc dissected immediately following the downshift (0 hr, I) and after 24 hr of regeneration (24 hr, J). (K) A control disc for notum ablation (*pnr^{ts}>GFP*). (L-M) *pnr^{ts}>GluR1,GFP* ablated discs at 0 hr (L) and 24 hr (M), open arrowheads highlight the absence of the Wg notum stripe following ablation. (N) A control disc for hinge ablation (*R73G07^{ts}>GFP*). (O-P) *R73G07^{ts}>GluR1,GFP* ablated discs at 0 hr (O) and 24 hr (P). Open arrowhead in (O) indicates the absence of NiA in the adjacent wing pouch. (Q) A control disc for posterior compartment ablation (*hh^{ts}>GFP*). (R-S') *hh^{ts}>GluR1,GFP* ablated discs at 0 hr (R-R') and 24 hr (S-S'). Dotted lines in (R') and (S') show an absence of NiA in the anterior hinge. (T) A quantification of the NiA response in the anterior pouch, hinge, and notum in control (n=13), 0 hr (n=14), and 24 hr (n=12) discs in response to *hh* ablation. Error bars represent the standard deviation. NiA are defined as cDcp-1-positive (cDcp-1+ve), GFP-negative (GFP-ve) cells. See Supplementary Genotypes file for exact genotypes.

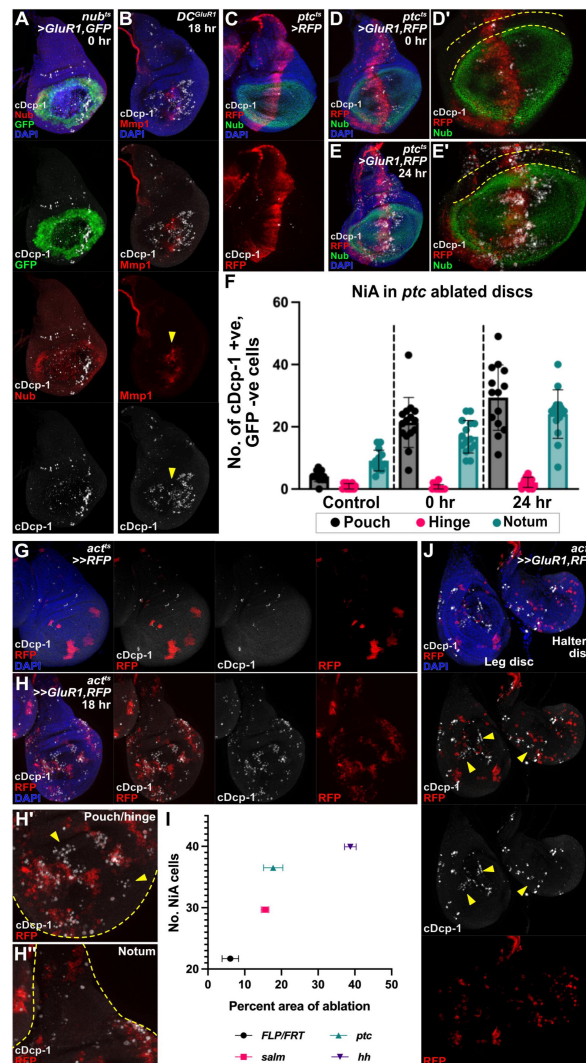


Figure 1 – figure supplement 1.

(A) A *nub^{ts}>GluR1,GFP* ablated disc at 0 hr. (B) A *DC^{GluR1}* ablated disc highlighting the overlap between wound edge apoptosis and the JNK target Mmp1 (arrowhead) and the absence of JNK signaling in NiA cells. (C) A control disc for anterior compartment ablation (*ptc^{ts}>RFP*). (D-E') *ptc^{ts}>GluR1,RFP* ablated discs at 0 hr (D-D') and 24 hr (E-E'). Dotted lines in (D') and (E') highlight the lack of NiA cells in the hinge. (F) A quantification of the NiA response to *ptc* ablation in the pouch, hinge, and notum of control (n=14), 0 hr (n=15), and 24 hr (n=14) discs, with NiA being defined as cDcp-1-positive (+ve), RFP-negative (-ve) cells. Error bars represent the standard deviation. (G) A control disc for *FLP/FRT* clonal ablation, where actin "flipout" GAL4 (*act^{ts}>>RFP*) drives the expression of *UAS-RFP* in clonal patches of cells after a short heat shock (see materials and methods for details). (H-H'') A wing disc with clonal patches of *UAS-RFP*, *UAS-GluR1* expressing cells. (H') is a close-up of pouch and hinge clones with arrowheads pointing to NiA cells, while (H'') is a close-up of notum clones. (I) A quantification of the number of NiA cells induced following ablation with *act^{ts}>>GluR1* (n=11), *R85E08^{ts}>GluR1* (n=10), *ptc^{ts}>GluR1* (n=15), and *hh^{ts}>GluR1* (n=14), highlighting that NiA number increases with increasing area of ablation. (J) A leg and a haltere disc with clonal patches of *UAS-RFP*, *UAS-GluR1* expressing cells. Small patches of cDcp-1+ve, RFP-ve cells in both discs suggests that the NiA response may occur in other imaginal discs and is not limited to the wing disc. See Supplementary Genotypes file for exact genotypes.

the relevant DAMPs or PRRs may also be responsible for the pattern of NiA formation throughout the disc. Alternatively, there may be inhibitory factors present in notum cells that protect against the induction of pro-apoptotic pathway elements. Indeed, differences in caspase 3 activation within the notum and pouch have been shown (Bergantino *et al.* 2010). Thus, these data suggest that either only the pouch releases DAMPs - and has the requisite PRRs to respond to these DAMPs - that lead to NiA following necrosis, or that unknown genetic factors limit the capacity of the notum and hinge to respond to damage.

As the efficacy of DAMPs might be limited by how far they can reach after being released from lysed cells, we also induced necrosis in the entire posterior disc compartment with *hedgehog-GAL4* (*hh^{ts}>GluR1*) (Figure 1Q), and in an anterior stripe along the anterior/posterior compartment boundary using *patched-GAL4* (*ptc^{ts}>GluR1*) (Figure 1 – figure supplement 1C). These experiments cause the simultaneous necrosis of pouch, hinge and notum tissues, allowing us to determine the potential of these different tissue identities to produce NiA. In both experiments, NiA cells are observed in the pouch, and to a lesser extent the notum, but are still strikingly absent from the hinge (Figure 1R, R' and T, Figure 1 – figure supplement 1D, D' and F). After 24 hr of recovery, there is an increase in the number of NiA within the pouch and notum, but not the hinge (Figure 1S, S' and T, Figure 1 – figure supplement 1E, E' and F). Thus, it appears that NiA can occur outside of the pouch when a large enough area, or an area that also includes the pouch, is ablated. However, the hinge remains completely refractory to NiA formation.

To avoid any bias in the use of tissue-specific *GAL4* drivers, we also made RFP-labeled stochastic clones that have the potential to undergo necrosis upon changing the growth temperature to 30°C (Figure 1 – figure supplement 1G). After allowing these clones to develop, we triggered necrosis in early third larval instar and examined the extent of active caspase in the different disc regions (Figure 1 – figure supplement 1H). As expected, necrosis of clones in the pouch leads to active caspases both within and surrounding the ablated area, including cells without the RFP clone label, suggesting that NiA has occurred (Figure 1 – figure supplement 1H', arrowheads). We also found that necrosis in the notum leads to comparatively little caspase labeling (Figure 1 – figure supplement 1H'') consistent with the notum being less able to generate NiA. Necrotic clones in the hinge also produces caspase activity, but most of these cells also have RFP, suggesting again that NiA does not occur in the hinge (Figure 1 – figure supplement 1H'). Using this approach we also examined whether NiA formation occurs in other larval tissues as well, and found that necrotic clones in both haltere and leg discs appear to result in an NiA response (Figure 1 – figure supplement 1J). Of note, the use of clones that naturally vary in size also demonstrates that the area of ablation is related to the amount of NiA produced (Figure 1 – figure supplement 1I), which is also true of our tissue-specific ablation experiments (Figure 1 – figure supplement 1I).

Together, these data infer three important conclusions: 1) all areas of the disc can be killed by necrosis and therefore potentially can release DAMPs, 2) NiA is limited to the pouch when local necrosis occurs, but when multiple (or large) areas of the disc are killed, limited NiA can also be induced in the notum, although we cannot rule out that this is due to DAMPs from dying pouch cells, and 3) the hinge is refractory to NiA, which is consistent with other findings that show its resistance to apoptosis in response to irradiation (Verghese and Su 2016). Thus, the overall pattern of competence to undergo NiA appears to reflect the uneven regenerative capacity of the wing disc, with NiA formation predominantly associated with the highly regenerative wing pouch.

NiA is regulated by WNT and JAK/STAT signaling

As NiA readily occurs in the pouch but is excluded from the nearby hinge, we used this contrasting response to identify genetic factors that might regulate NiA formation. The wing hinge is specified by JAK/STAT signaling during disc development, which can protect cells from irradiation-induced apoptosis potentially via the expression of *Zn finger homeodomain 2* (*zfh2*) (La Fortezza *et al.* 2016, Verghese and Su 2016, Verghese and Su 2018). Alongside JAK/STAT, the presence of

Wingless (Wg, *Drosophila* Wnt1), which encircles the pouch, may also protect cells from death and permit regeneration of the pouch through the repression of *reaper* (*rpr*) (Verghese and Su 2016). As such, we investigated both JAK/STAT and Wg to determine whether they regulate NiA formation.

The activity of the JAK/STAT pathway can be visualized in the hinge of early third instar larval discs by a *10XSTAT-GFP* reporter (Bach *et al.* 2007)(Figure 2A). Upon ablation with *DC^{GluR1}*, high levels of JAK/STAT activity are observed in the pouch at the immediate WE (Figure 2B, arrowhead), similar to its upregulation following irradiation or apoptotic ablation (Herrera and Bach 2019). As JNK signaling is induced at the WE (Figure 1 – figure supplement 1B) (Klemm *et al.* 2021), and the *unpaired* ligands are targets of JNK signaling (Katsuyama *et al.* 2015, Jaiswal *et al.* 2023), this JAK/STAT activity is likely to be JNK-mediated. By contrast, low levels of JAK/STAT activity are observed in the areas of the pouch where NiA occurs (Figure 2B, open arrowheads), surrounded by the higher developmental JAK/STAT in the hinge (Figure 2B). To determine if low JAK/STAT activity is important for NiA, we knocked down the receptor *domeless* (*UAS-dome^{RNAi}*) in the pouch, which results in a significant increase in NiA (Figure 2C, D and E), but has no effect when expressed without damage (Figure 2 – figure supplement 1A), suggesting that JAK/STAT signaling may negatively regulate the formation of NiA. To support this finding, we next ectopically activated JAK/STAT (*UAS-hop48A*). However, even in the absence of damage, this expression results in high levels of caspase positive cells (Figure 2 – figure supplement 1B and C), making it difficult to determine an effect on NiA formation. Therefore, to further investigate if JAK/STAT regulates NiA formation, we asked whether reducing developmental JAK/STAT in the hinge might lead to NiA spreading further into this region. We generated a version of DUAL Control that expresses *GAL4* in the posterior compartment by replacing the pouch-specific *DVE>>GAL4* with *hh-GAL4* (Figure 2F and G, Figure 2 – figure supplement 1D). To prevent *hh-GAL4* from being active throughout development, we included *GAL80^{ts}* (hereafter *DC^{GluR1}hh^{ts}*) and used temperature changes to limit *GAL4* activity to the period just prior to ablation (Figure 2F). With this system, we knocked down the expression of the JAK/STAT transcription factor *Stat92E* (*UAS-Stat92E^{RNAi}*) in the posterior compartment (Figure 2 – figure supplement 1E and F) and ablated the distal pouch, which again shows an increase in caspase-positive cells in the pouch (Figure 2H, arrowhead, quantified in Figure 2L), but surprisingly NiA cells are still not observed in the hinge (Figure 2H, open arrowhead). To further test this result, we also targeted *zinc finger homeodomain 2* (*Zfh2*), a downstream target of the JAK/STAT pathway that potentially protects cells from apoptosis (La Fortezza *et al.* 2016, Verghese and Su 2018). The knockdown of *Zfh2* slightly increases the formation of NiA, although not significantly (Figure 2 – figure supplement 1I and J, quantified in Figure 2L). However, we noted that the level of *Zfh2* protein did not appear strongly affected by this knockdown (Figure 2 – figure supplement 1I), and therefore we cannot confidently conclude that *Zfh2* is involved in caspase activation in the context of necrotic damage.

Expression of these RNAi lines targeting *Stat92E* or *Zfh2* under non-ablating conditions does not yield any increase in caspase signal (Figure 2 – figure supplement 1G and I). Thus, JAK/STAT signaling appears to limit NiA formation in the pouch, while the inability for NiA to expand into the hinge upon reducing JAK/STAT suggests that other hinge-specific factors may be involved.

Wg has also been shown to protect cells from apoptosis in the hinge (Verghese and Su 2016), and therefore could influence the formation of NiA. Unlike the stochastic temperature change-induced apoptosis (Figure 2I), we noted that NiA in the pouch frequently occurs in discrete populations that avoid Wg at the margin stripe and the inner Wg circle at the boundary of the pouch and hinge (Figure 2J), limiting formation of the NiA cells to regions that appear to overlap the *vestigial quadrant enhancer* (*vgQE-lacZ*, Figure 2 – figure supplement 1K and L) (Kim *et al.* 1996). By contrast, *cDcp-1*-positive cells at the WE do not avoid the Wg margin stripe (Figure 2J, arrowhead), suggesting that this behavior may be specific to NiA. To test this, we utilized *DC^{GluR1}hh^{ts}* to knock down *wg* in the posterior compartment of the disc (*UAS-wg^{RNAi}*) and

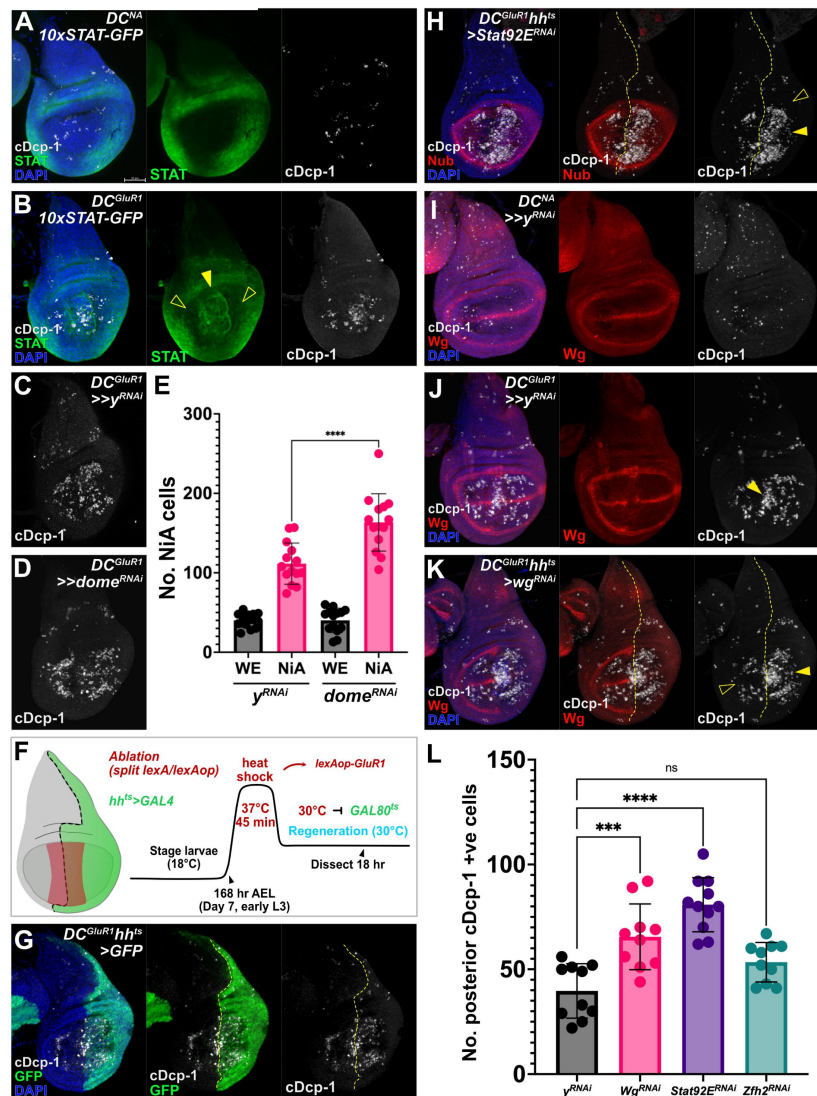


Figure 2.

NiA formation in pouch cells is regulated by the Wg and JAK/STAT pathways.

(A). A control disc (DC^{NA}) bearing the $10xSTAT-GFP$ reporter, showing the hinge-specific Stat92E activity that is normally absent from the wing pouch. (B) A DC^{GluR1} ablated disc bearing $10xSTAT-GFP$. Following damage, pouch-specific reporter expression is observed at high levels at the wound edge (arrowhead) and low levels at the NiA area of the pouch (open arrowheads). (C-E) $DC^{GluR1} >> y^{RNAi}$ (C, $n=14$) versus $DC^{GluR1} >> dome^{RNAi}$ (D, $n=14$) ablated discs, with the number of NiA quantified in (E), $P^{****} < 0.0001$. Data were analyzed with a one-way ANOVA followed by a multiple comparisons test. Error bars represent the standard deviation. (F) A $DC^{GluR1} hh^{ts} > GAL4$ ablation schematic ($DC^{GluR1} hh^{ts}$). A short heat shock (45 min at $37^{\circ}C$) induces split $LexA/lexAop-GluR1$ ablation, while a heat shift (20 hr at $30^{\circ}C$) following the heat shock will inactivate $GAL80^{ts}$ and permit $UAS-X$ expression. (G) A $DC^{GluR1} hh^{ts} > GFP$ ablated disc, where GFP highlights the area of the disc being targeted for JAK/STAT and Wg knockdown in the following experiments. The posterior NiA will be assayed in response to knockdown while anterior NiA serve as an internal control. (H) A $DC^{GluR1} hh^{ts} > Stat92E^{RNAi}$ ablated disc, arrowhead highlights an increase in pouch NiA, open arrowhead indicates the lack of cDcp-1 signal in the hinge. (I) A control DC^{NA} disc showing developmental Wg expression. (J) A $DC^{GluR1} >> y^{RNAi}$ ablated disc showing that NiA cells avoid the areas of the pouch with high Wg expression, while wound edge apoptotic cells overlap the developmental and damage-specific Wg-expressing cells (arrowhead). (K) A $DC^{GluR1} hh^{ts} > wg^{RNAi}$ ablated disc with NiA cells observed at the posterior margin (arrowhead) while anterior NiA avoid the Wg margin stripe (open arrowhead). (L) A quantification of posterior NiA cells (cDcp-1+ve cells within the pouch) following a control knockdown (y^{RNAi} , $n=10$), $Stat92E^{RNAi}$ ($n=10$), wg^{RNAi} ($n=10$), and $Zfh2^{RNAi}$ ($n=11$). $P^{***} = 0.0002$, $P^{****} < 0.0001$. ns = not significant. Data were analyzed with a one-way ANOVA followed by a multiple comparisons test. Error bars represent the standard deviation. See Supplementary Genotypes file for exact genotypes.

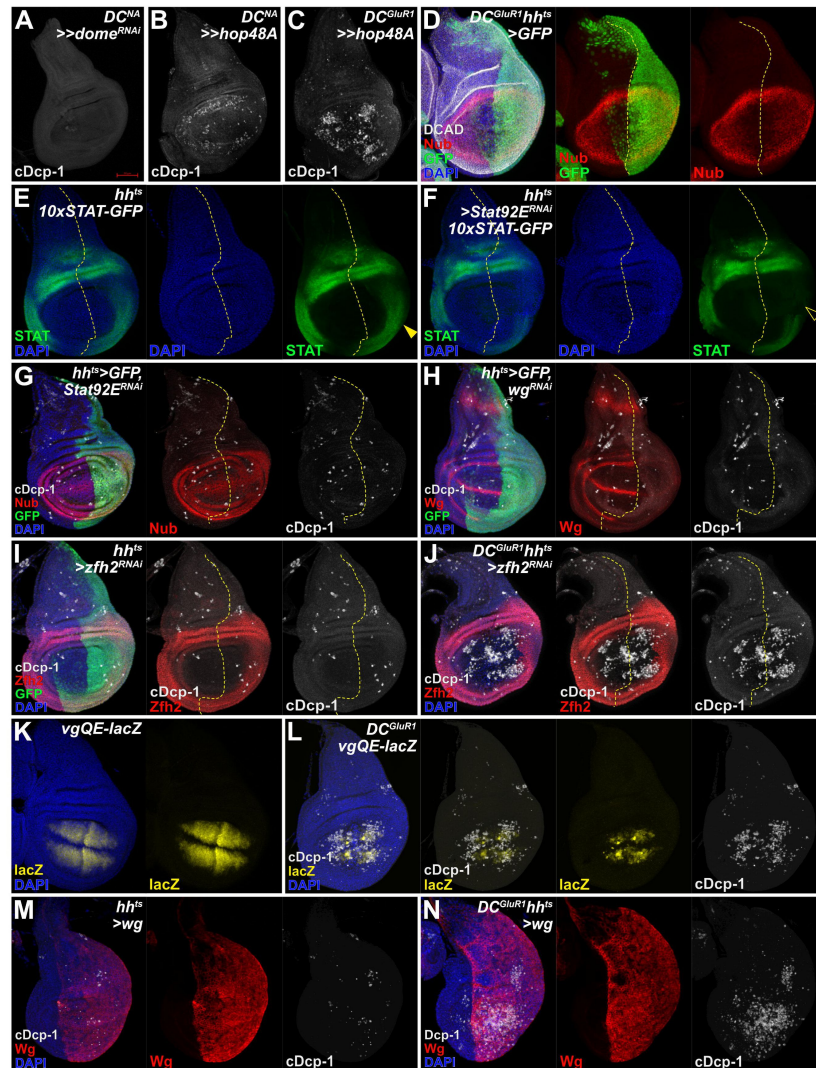


Figure 2 – figure supplement 1.

(A) $DC^{NA}>>dome^{RNAi}$ control disc with no apoptosis in the absence of ablation. (B) A $DC^{NA}>>hop48A$ control disc with high levels of cDcp-1 observed throughout the pouch. (C) A $DC^{GluR1}>>hop48A$ ablated disc with high levels of apoptosis throughout the pouch, which may be induced by $UAS-hop48A$ expression as seen in (B). (D) A $DC^{GluR1}hh^{ts}>GFP$ ablated disc with nubbin and DCAD highlighting the ablated *sal*m domain and GFP labeling the area of $UAS-X$ expression. (E) A control $hh-GAL4$ disc showing the normal expression pattern of the 10xSTAT-GFP reporter. (F) A $hh-GAL4>UAS-Stat92E^{RNAi}$ disc expressing 10xSTAT-GFP, demonstrating a reduction of JAK/STAT activity as revealed by the 10xSTAT-GFP reporter (arrowhead in H versus open arrowhead in I). (G) A control $hh^{ts}>GFP,Stat92E^{RNAi}$ disc showing no changes in cell death. (H) A control $hh^{ts}>GFP,wg^{RNAi}$ disc showing the absence of Wg in the posterior pouch and no change in posterior cell death. (I) A control disc (I) A $DC^{NA}>>zfh2^{RNAi}$ control disc showing no increased apoptosis in the absence of ablation. (J) A $DC^{GluR1}hh^{ts}>zfh2^{RNAi}$ ablated disc showing a slight increase in pouch NiA. (K) A control disc bearing the *vgQE-lacZ* reporter showing the area of the pouch with vestigial quadrant enhancer expression. (L) A DC^{GluR1} ablated disc bearing *vgQE-lacZ*, showing the high degree of overlap between NiA cells and lacZ-expressing cells. (M) A control $hh^{ts}>wg$ disc with no strong changes in cell death observed. (N) A $DC^{GluR1}hh^{ts}>wg$ ablated disc showing an increase in apoptosis across the wing pouch and in the posterior pleura. See Supplementary Genotypes file for exact genotypes.

found that NiA now occurs in areas of the pouch where *wg* expression is lost, including the wing margin and inner hinge (**Figure 2K**, arrowhead, quantified in **Figure 2L**), unlike NiA in the anterior (**Figure 2K**, open arrowhead). The increase in cDcp-1-positive cells does not occur when *wg* signaling is similarly blocked without damage (**Figure 2 – Figure supplement 1H**). Notably, the converse experiment in which *wg* is ectopically expressed during ablation does not suppress NiA (**Figure 2 – Figure supplement 1M** and **N**), consistent with our hypothesis that other factors, such as targets downstream of JAK/STAT, might act alongside Wg to regulate NiA. Together, these data demonstrate that both WNT and JAK/STAT signaling act to limit NiA, thus potentially constraining it to the pouch following necrosis.

NiA promotes proliferation late in regeneration

As NiA is spatially regulated by at least two major signaling pathways in the disc, we next focused on how the localization of NiA relates to its role in promoting regeneration. In our previous work, using an E2F reporter (*PCNA-GFP*) we found that the appearance of NiA coincides with localized proliferation in the distal pouch at 18 hr post-ablation close to the wound, which persists at 24 hr post ablation (Klemm *et al.* 2021).

However, investigation at subsequent time points of recovery shows that regenerative proliferation continues to increase through 36 hr and 48 hr of regeneration, later than we initially assayed, with a significant increase in E2F reporter expression occurring between 18 and 36 hr in the pouch relative to the whole disc, (**Figure 3A-A**, **Figure 3** – Figure supplement 1A and B, quantified in C). To investigate this proliferative response, we used EdU to assay the relative level of cell proliferation in discs throughout regeneration from 18 hr to 48 hr post-ablation with *DC^{GluR1}* (**Figure 3B-B**), using folds as landmarks to normalize EdU intensity in the pouch relative to the disc (**Figure 3 – figure supplement 1D-D**). We also performed the same time course with apoptotic ablation using *DC^{hepCA}* for comparison (**Figure 3C-C**). At 18 hr of regeneration following ablation with either *DC^{GluR1}* or *DC^{hepCA}*, cells at the WE have already migrated distally to close the injury, while EdU is absent from an area immediately adjacent to the wound (**Figure 3B** and **C**). This is consistent with the recently described JNK-mediated pause in proliferation that occurs in regenerating wing discs (Jaiswal *et al.* 2023). At 24 hr, this proliferation-devoid area continues to persist following apoptotic injury (*DC^{hepCA}*) and the EdU signal becomes elevated broadly across the rest of the pouch, representing the formation of a blastema (**Figure 3C**). By contrast, at 24 hr after necrosis (*DC^{GluR1}*), the EdU label is reestablished in cells around the wound showing that proliferation has restarted in these cells (**Figure 3B**).

Unlike *DC^{hepCA}*, the rest of the pouch does not appear to change its rate of proliferation (**Figure 3B**). By 36 hr following necrosis, a higher EdU signal occurs broadly across the pouch relative to the rest of the disc, which is significantly stronger compared to discs ablated by apoptosis (**Figure 3B**, C' and E). This localization is maintained at 48 hr and remains consistently higher in *DC^{GluR1}* versus *DC^{hepCA}* ablated discs (**Figure 3B**, C'' and E). Thus, the timeline of recovery from necrosis appears to be distinct from that of apoptotic injury, with the strongest changes in regenerative proliferation occurring at comparatively later stages.

The localized proliferation at 36 hr occurs after the appearance of NiA. We previously showed that blocking the apoptotic pathway by simultaneously knocking down DIAP1 inhibitors *rpr*, *hid* and *grim* (*UAS-mir(RHG)*, (Siegrist *et al.* 2010)) throughout the pouch limits the initial change in proliferation at early stages (18 – 24 hr) and inhibits regeneration (Klemm *et al.* 2021). However, it remains unclear whether this newly observed localized proliferation at 36 hr and 48 hr also relies on a functional apoptotic pathway, and moreover, to what extent this regenerative proliferation relies on the JNK-mediated apoptosis at the WE versus the JNK-independent NiA in the LP. To answer these questions, we blocked apoptosis throughout the pouch and this time examined proliferation in late regeneration using EdU (**Figure 3D-D**). We found that the localized increase in EdU signaling at 36 hr is lost (**Figure 3D** and **F**), although by 48 hr this

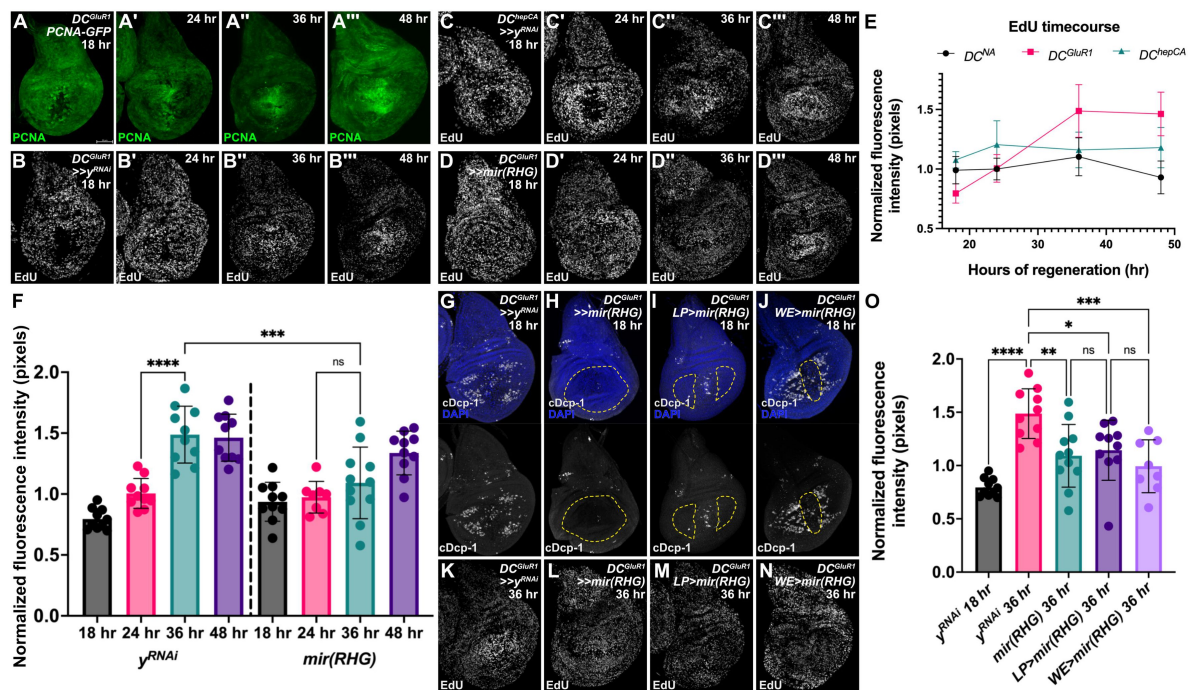


Figure 3.

NiA promotes proliferation late in regeneration.

(A-A''') A time course of DC^{GluR1} ablated discs bearing the $PCNA-GFP$ reporter at 18 hr (A), 24 hr (A'), 36 hr (A''), and 48 hr (A'''). (B-B''') A time course of EdU in $DC^{GluR1} >> y^{RNAi}$ ablated discs at 18 hr (B), 24 hr (B'), 36 hr (B''), and 48 hr (B'''). Over time, high EdU signal localizes to the damaged pouch by 36 hr and remains elevated at 48 hr. (C-C''') A time course of $DC^{hepCA} >> y^{RNAi}$ ablated discs at 18 hr (C), 24 hr (C'), 36 hr (C''), and 48 hr (C'''). (D-D'''), demonstrating that EdU initially becomes elevated in the outer regions of the pouch by 24, but then localizes to the central pouch region by 36 hr. A time course of $DC^{GluR1} >> mir(RHG)$ ablated discs at 18 hr (D), 24 hr (D'), 36 hr (D''), and 48 hr (D'''), demonstrating the loss of EdU localization. (E) A graph of the DC^{GluR1} , DC^{hepCA} , and DC^{NA} EdU time courses highlight the pattern of EdU labeling in the wing pouch between each system. Error bars represent the standard deviation (F) A quantification of EdU signal intensity between $DC^{GluR1} >> y^{RNAi}$ at 18 hr (n = 10), 24 hr (n = 10), 36 hr (n = 10), 48 hr (n = 10), and $DC^{GluR1} >> mir(RHG)$ at 18 hr (n = 10), 24 hr (n = 8), 36 hr (n = 11), and 48 hr (n = 10) time courses, ns, not significant; ****P = 0.0002; *****P < 0.0001; data were analyzed with a one-way ANOVA and multiple comparisons test. Error bars represent the standard deviation. (G-J) DC^{GluR1} ablated discs with different populations of apoptotic cells suppressed by $mir(RHG)$. (G) A control $DC^{GluR1} >> y^{RNAi}$ ablated disc showing the typical pattern of NiA formation. (H) A $DC^{GluR1} >> mir(RHG)$ ablated disc suppressing both wound edge apoptosis and NiA cells. (I) A $DC^{GluR1} >> mir(RHG); DR^{WNT}-GAL80$ ablated disc, which targets the NiA area of the pouch for $UAS-mir(RHG)$ expression (hereafter Lateral Pouch, or $LP > mir(RHG)$). (J) A $DC^{GluR1} \times R85E08 > mir(RHG)$ ablated disc, which targets wound edge apoptotic cells for suppression (hereafter Wound Edge, or $WE > mir(RHG)$). The dotted lines each panel highlight the area of $UAS-mir(RHG)$ expression. (K-N) DC^{GluR1} ablated discs with representative EdU labels at R36 in response to y^{RNAi} (K), whole-pouch $mir(RHG)$ (L), $LP > mir(RHG)$ (M), and $WE > mir(RHG)$ (N). (O) A quantification of the normalized EdU fluorescent intensity of $DC^{GluR1} >> y^{RNAi}$ R18 (n = 10), $DC^{GluR1} >> y^{RNAi}$ R36 (n = 10), $DC^{GluR1} >> mir(RHG)$ R36 (n = 11), $LP > mir(RHG)$ (n = 10), and $WE > mir(RHG)$ (n = 9) ablated discs; *P = 0.0189, **P = 0.0044, ***P = 0.0008, *****P < 0.0001; data were analyzed with a one-way ANOVA and multiple comparisons tests. Error bars represent the standard deviation. LP = lateral pouch, WE = wound edge. See Supplementary Genotypes file for exact genotypes.

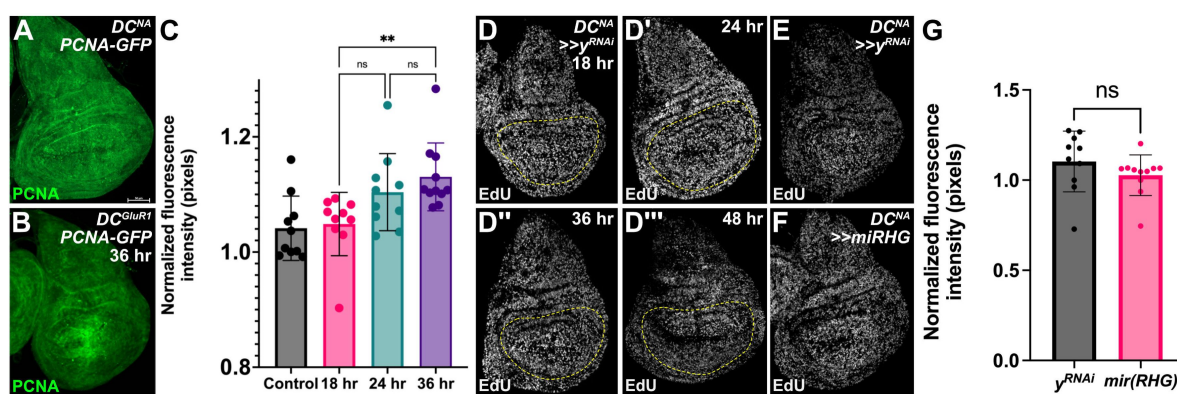


Figure 3 – figure supplement 1.

(A) A DC^{NA} control disc expressing PCNA-GFP. (B) A DC^{GluR1} ablated disc expressing PCNA-GFP, with reporter expression localizing to the damaged pouch over time. (C) A quantification of DC^{NA} (n = 10) and DC^{GluR1} ablated discs at 18 hr (n = 10), 24 hr (n = 10), and 36 hr (n = 11), with the pouch-specific PCNA-GFP signal normalized to entire discs in each time point. This time course demonstrates that PCNA-GFP expression localizes to the distal pouch by 36 hr of regeneration. Data were analyzed with a one-way ANOVA followed by a multiple comparisons test. $P^{**} = 0.0095$. ns = not significant. Error bars represent the standard deviation. (D-D''') A $DC^{NA} >> y^{RNAi}$ control time course at 18 hr (D), 24 hr (D'), 36 hr (D''), and 48 hr (D'''). The folds of the wing pouch, visualized by DAPI, were used to outline the EdU signal for quantification in all experiments. (E-F) A $DC^{NA} >> y^{RNAi}$ control disc (E) with EdU labeling relative to a $DC^{NA} >> mir(RHG)$ control disc (F). (G) A quantification of discs in (E, n=10) and (F, n=10) demonstrating that miRHG expression does not alter EdU levels, ns, not significant. Data were analyzed with a Student's T-test. See Supplementary Genotypes file for exact genotypes.

increase is mostly restored (**Figure 3D** and F). Importantly, the expression of *mir(RHG)* does not influence EdU levels in the absence of damage (**Figure 3 – figure supplement 1E** and **G**). Together, these data confirm that a functional apoptotic pathway is necessary to induce a localized proliferative response late in regeneration following necrosis. To understand the relative contribution of WE apoptosis or the NiA in the LP, we designed experiments to block apoptosis in each disc area alone (*LP>mir(RHG)*, **Figure 3I**, and *WE>mir(RHG)*, **Figure 3J**, see Materials and Methods for genotypes) relative to the whole pouch knock down (**Figure 3G** and **H**). Strikingly, the high levels of EdU normally present at 36 hr are strongly reduced when apoptosis is blocked in either population (**Figure 3K-O**), suggesting that both dying cells at the WE and the NiA in the LP contribute to regenerative proliferation following necrosis. These data agree with our previous findings that both populations are necessary for the overall ability to regenerate adult wings (Klemm *et al.* 2021).

NiA does not promote proliferation through Apoptosis-induced Proliferation (AiP)

The question remains as to how NiA promotes regenerative proliferation. In *Drosophila*, cells undergoing apoptosis secrete factors such as Wg and Dpp to induce the proliferation of neighboring cells as part of a JNK-dependent Apoptosis-induced Proliferation (AiP) (Fogarty and Bergmann 2017). Although NiA occurs independent of JNK, to determine whether NiA-induced proliferation relies on any of the same signaling factors as AiP, we examined the damage-specific expression of these various secreted factors. To ensure we could visualize such signals, we used *lacZ*-based reporters and generated undead cells by expressing the baculoviral P35 (*UAS-P35*) in the whole pouch. This protein inhibits activity of the effector caspases Drice and cDcp-1 to block cell death (Hawkins *et al.* 2000, Meier *et al.* 2000), thus allowing signals produced by these cells to be readily detected. Following ablation with *DC^{GluR1}*, ectopic *wg* and *dpp* expression (*wg-lacZ* and *dpp-lacZ*) is observed at the WE (**Figure 4C and C'** versus A-B', and **Figure 4F** and **F'** versus D-E', arrowheads in C' and F') coinciding with JNK activity in this region (Klemm *et al.* 2021). However, *lacZ* is not observed in the LP where NiA occurs (**Figure 4C** and F', open arrowheads), indicating that these cells do not activate these mitogens. Similarly, we did not see expression of the EGF ligand *spitz* (*spi-lacZ*) in *DC^{GluR1}* ablated discs (**Figure 4 – figure supplement 1A** and **B**), which is observed during AiP in the eye (Fan *et al.* 2014). These results suggest that NiA does not promote proliferation through the same signaling factors as those seen during AiP.

We also tested whether other elements required for AiP are involved in NiA-induced proliferation. In addition to mitogen production, AiP also involves the production of extracellular reactive oxygen species (ROS) through a non-apoptotic function of Dronc that activates Duox. (Fogarty *et al.* 2016, Fogarty and Bergmann 2017, Amcheslavsky *et al.* 2018, Diwanji and Bergmann 2018). We first examined the extent of ROS production using dihydroethidium (DHE). This assay showed high levels of ROS localized to the WE but not in the LP (**Figure 4G**, arrowhead), suggesting that NiA does not produce ROS during regeneration.

Consistent with this finding, the removal of ROS through pouch-wide expression of either the ROS chelators *Catalase* and *Superoxide dismutase 1* (*UAS-Cat*; *UAS-Sod1*) or knockdown of *Duox* (*UAS-Duox^{RNAi}*) has no observable effect on the appearance of NiA (**Figure 4H** and **I**), although apoptosis at the WE is strongly suppressed in both experiments (**Figure 4H** and **I**, open arrowhead). It has also been shown that the Duox maturation factor *moladietz* (*mol*) is upregulated following injury to sustain the production of ROS (Khan *et al.* 2017, Pinal *et al.* 2018). However, while a minor increase in *lacZ* expression occurs at the WE (**Figure 4 – figure supplement 1C** and **D**, arrowhead), no change in the expression of a *mol* reporter (*mol-lacZ*) is observed in response to necrosis in the LP. Finally, to functionally test whether AiP is required for the proliferation associated with NiA, we examined EdU levels across the disc at 36 hr when P35 is expressed. Normally, when undead cells are created via P35, ectopic mitogen production

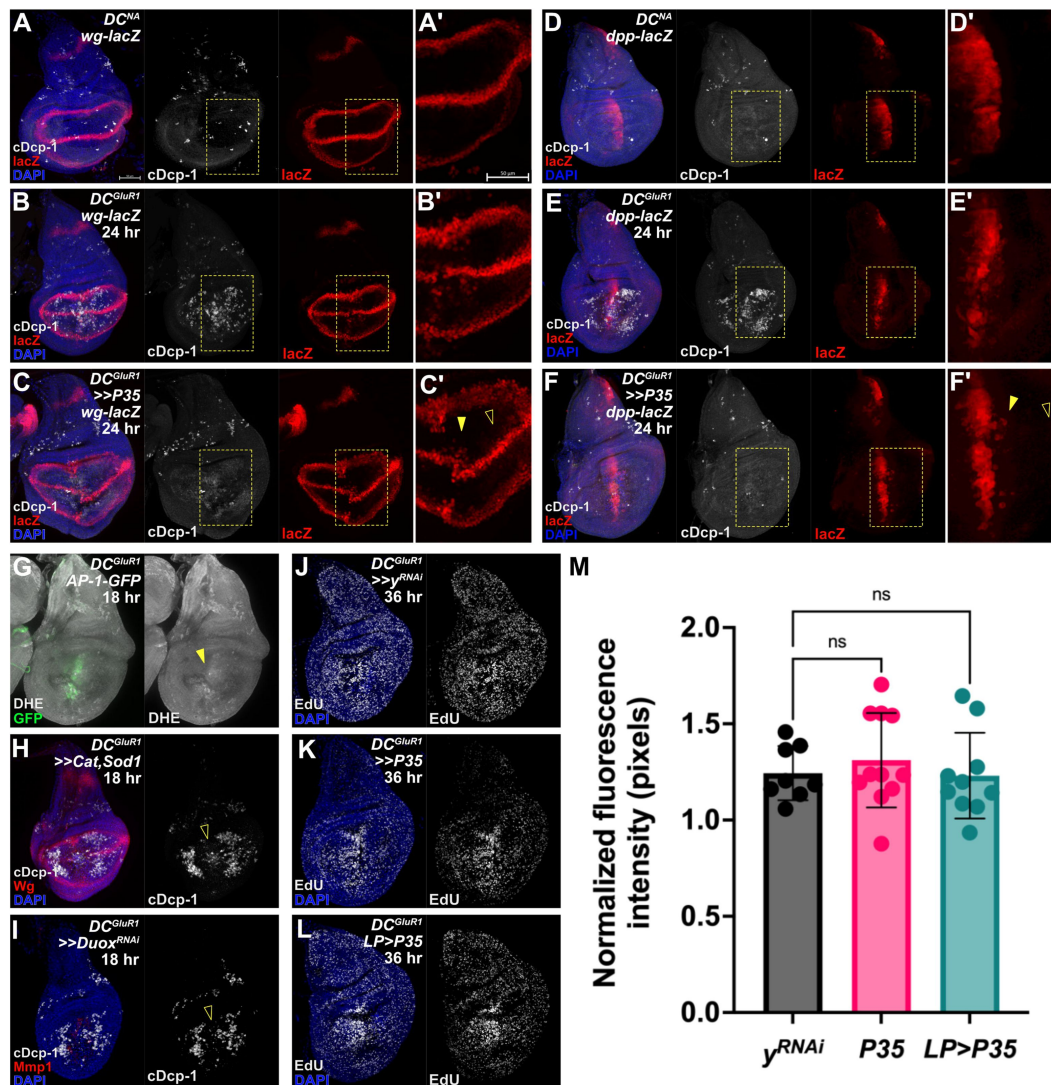


Figure 4

NiA does not secrete mitogens associated with Apoptosis-induced Proliferation (AiP).

(A-A') A DC^{NA} control disc bearing the *wg-lacZ* reporter. (B-B') A DC^{GluR1} ablated disc with *wg-lacZ* showing the expression of the reporter during regeneration, note several *lacZ*-expressing cells at the wound edge. The 24 hr time point was chosen as an intermediate between NiA formation (18 hr) and blastema formation (36 hr), as any secreted factors involved in AiP are likely to be detected prior to 36 hr. (C-C') A $DC^{GluR1} >> P35$ ablated disc bearing *wg-lacZ* with *lacZ* expressing cells observed at the wound edge (arrowhead) but not in the lateral pouch where NiA occur (open arrowhead). (D-D') A DC^{NA} control disc bearing the *dpp-lacZ* reporter. (E-E') A DC^{GluR1} ablated disc with *dpp-lacZ*; *lacZ*-expressing cells are again observed at the wound edge. (F-F') A $DC^{GluR1} >> P35$ ablated disc bearing *dpp-lacZ*, with *lacZ*-expressing cells at the wound edge (arrowhead) but not where NiA cells occur (open arrowhead), demonstrating that NiA do not secrete factors (Dpp and Wg) involved in AiP. (G) A DC^{GluR1} ablated disc bearing the *AP-1-GFP* reporter and labeled with dihydroethidium (DHE) to visualize reactive oxygen species (ROS). Low levels of DHE labeling are observed at the wound edge overlapping *AP-1-GFP* expression (arrowhead), while no DHE labeling is observed in the lateral pouch where NiA occur. (H) A $DC^{GluR1} >> Cat, Sod1$ ablated disc showing reduced wound edge apoptosis (open arrowhead) but no change in NiA formation. (I) A $DC^{GluR1} >> Duox^{RNAi}$ ablated disc showing a similar pattern to (H) with an observed loss of wound edge apoptosis (open arrowhead) but no change in NiA formation. (J-L) DC^{GluR1} ablated discs bearing *yRNAi* (J), *P35* (K), and *LP>P35* (L) at 36 hr and labeled with EdU. (M) A quantification of discs in (J-L) demonstrate no change in EdU labeling upon the expression of *P35* in the whole pouch (K) or in NiA cells (*LP>P35*, L), confirming that NiA promote proliferation independent of AiP, (*y^{RNAi}*, $n=8$, *P35*, $n=11$, *LP>P35*, $n=10$), ns, not significant. Data were analyzed with a one-way ANOVA followed by a multiple comparisons test. LP = lateral pouch, AiP = apoptosis-induced proliferation. See Supplementary Genotypes file for exact genotypes.

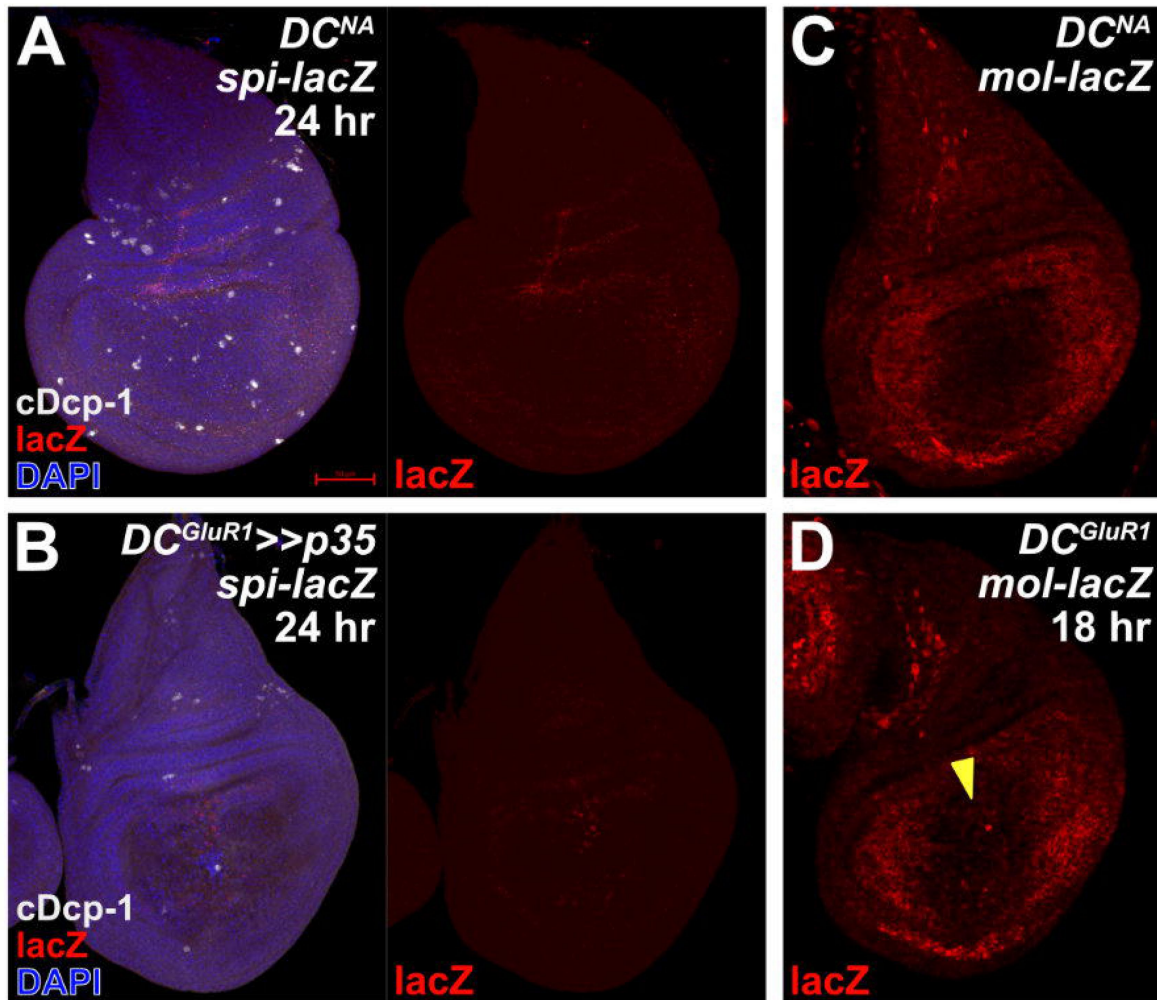


Figure 4 – figure supplement 1.

(A) A DC^{NA} control disc bearing the *spi-lacZ* reporter, showing no *lacZ* in the wing disc. (B) A $DC^{GluR1} >> p35$ ablated disc bearing *spi-lacZ* showing few *lacZ*-expressing cells at the wound edge. (C) A DC^{NA} control disc bearing *mol-lacZ* highlighting the developmental expression pattern of *mol* in the wing disc. (D) A DC^{GluR1} ablated disc bearing *mol-lacZ*. A slight increase in *lacZ* expression is observed at the wound edge (arrowhead). See Supplementary Genotypes file for exact genotypes.

results increased proliferation and tumorous overgrowth. However, when P35 is expressed solely in the LP we saw no change in EdU labeling versus controls (**Figure 4J**, **L** and **M**), suggesting NiA do not form undead cells. When expressed in the whole pouch, P35 has a small but non-significant effect on EdU (**Figure 4J**, **K** and **M**), consistent with undead cells now being generated at the WE. Together, these data indicate that cells at the WE undergo AiP to contribute to regenerative proliferation, while NiA promote proliferation through a different mechanism.

A subset of cells undergoing NiA are both caspase-positive and have markers of DNA repair and proliferation

Robust populations of NiA appear in the wing pouch at around 18 hr of regeneration, (Klemm *et al.* 2021), while the localized change in EdU labeling that encompasses much of the damaged pouch is detected later in regeneration at 36 hr and 48 hr (**Figure 3B-B**, E and F). Since the loss of NiA abolishes this change in proliferation (**Figure 3F**, L-M and O), we sought to understand how NiA might be influencing proliferation at these later time points. To do so, we used a robust sensor for the activity of the effector caspases cDcp-1 and Drice called *Green Caspase-3 Activity indicator* (*UAS-GC3Ai*, (Schott *et al.* 2017)) to label NiA throughout regeneration (**Figure 5A-E**). GC3Ai consists of a cyclized GFP with the N and C termini linked by a short peptide that contains a caspase-3 cleavage site (DEVD), which prevents fluorescence under non-apoptotic conditions. Activated effector caspases can cleave the linker peptide, resulting in a fluorescent GFP signal that reports caspase activity in real time. Schott *et al.* demonstrated that apoptotic cells labeled by GC3Ai are extruded within 1 hr of labeling, consistent with previously published work on apoptotic cell extrusion in imaginal discs (Monier *et al.* 2015). With this reporter we found that, unlike normal apoptotic cells that are rapidly cleared from the wing disc, NiA persist and increase in abundance by 36 hr when proliferation localizes to the distal pouch (**Figure 5C**, quantified in **Figure 5F**), as well as at 48 hr (**Figure 5D**, quantified in **Figure 5F**), and even up to 64 hr (wandering stage), when regeneration is complete and pouch tissue is restored (**Figure 5E**, arrowhead). These persistent GC3Ai-positive cells are also frequently associated with adjacent cells actively undergoing mitosis by 36 hr, indicated by PH3 labelling (**Figure 5G-G**). Since the position of NiA cells change over time from 18 hr when they first appear to 36 hr when the change in proliferation occurs, to confirm that these GC3Ai-positive cells originally derive from NiA in the LP rather than the WE, we restricted GC3Ai expression to this part of the pouch (*LP>GC3Ai*, **Figure 5H** and **I**). The GC3Ai construct is tagged with an HA epitope that can be used to show its expression even in the absence of activation by caspases (**Figure 5 – figure supplement 1A** and **B**), which we used to confirm that its expression is limited to the LP (**Figure 5H** and **I**, **Figure 5 – figure supplement 1C**). With this experimental setup, we found that GC3Ai-positive cells are still present throughout the pouch at 36 hr, including the distal *sal* domain (**Figure 5I**, arrowhead, versus 5H, open arrowhead, quantified in **Figure 5 – Figure supplement 1D**), confirming that the source of persistent caspase-positive cells is indeed NiA rather than the WE.

As NiA cells appear to be maintained in the disc for an extended period, we wondered how their behavior and morphology compared to that of normal apoptotic cells. When apoptosis is induced in wing discs using *DC^{hepCA}*, GC3Ai labeling shows apoptotic cells present in the pouch at 18 hr (**Figure 5J**) throughout the disc proper (**Figure 5J**). At 36 hr these cells appear pyknotic and are basally extruded by 64 hr (**Figure 5K** and **K**). By comparison, upon ablation with *DC^{GluR1}*, caspase-positive cells are seen to occupy different regions of the disc at 18 hr, with WE apoptotic cells closer towards the basal surface and NiA derived from the LP still within the disc proper (**Figure 5L** and **L**). By 36 hr, the NiA form two distinct populations, some that are rounded up and appear closer to the basal surface, similar to the WE cells (**Figure 5M** and **M**, red arrowhead), and others that continue to exhibit a columnar appearance and contact both the apical and basal surfaces of the disc (**Figure 5M** and **M**, yellow arrowheads). The appearance and position of these cells suggest that a proportion of NiA cells may fail to complete apoptosis but instead persist into late stages of regeneration, despite the presence of detectable caspase activity. This is supported by the observation that only a minority of GC3Ai-positive cells have blebbing and

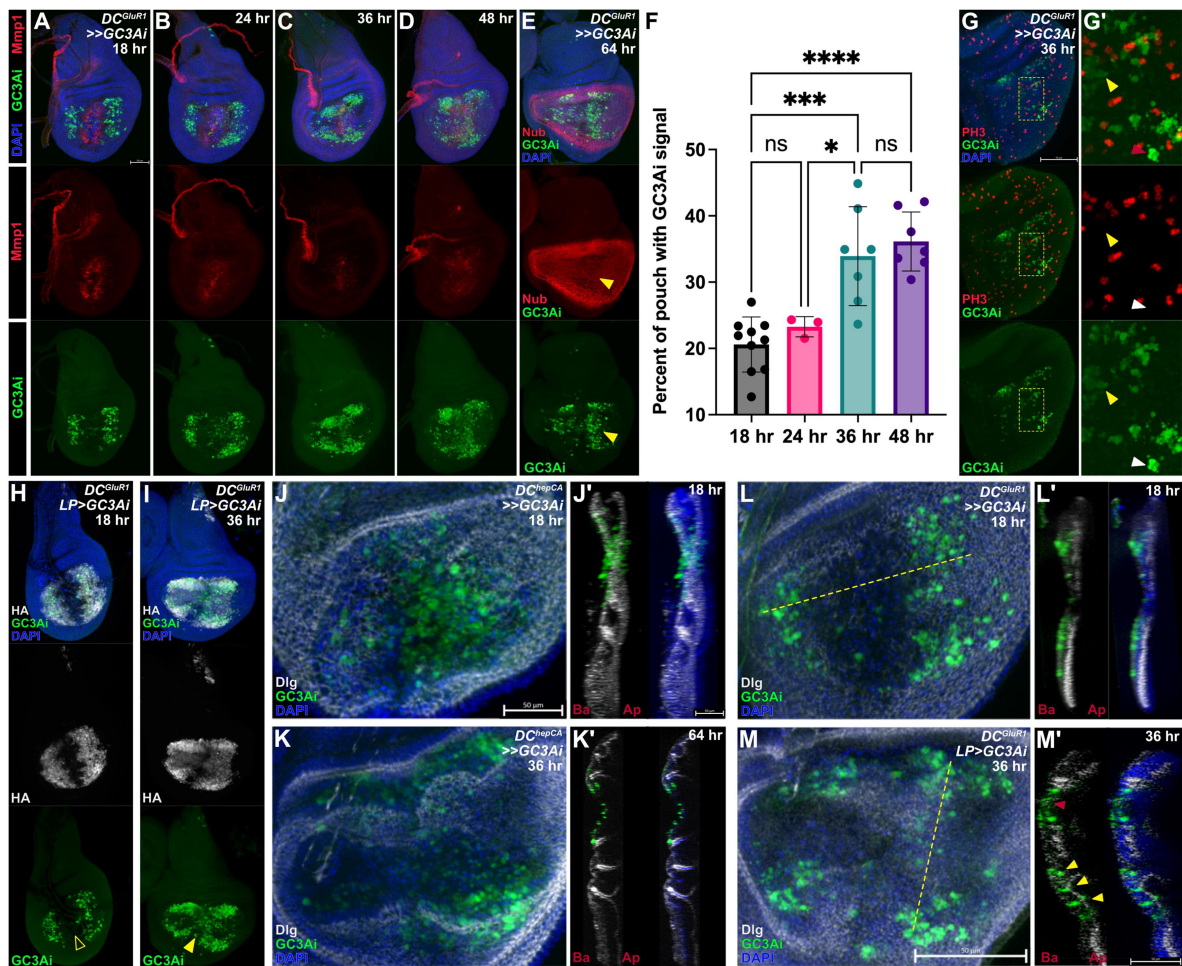


Figure 5.

NiCP cells persist in the disc proper throughout regeneration.

(A-D) A time course of *DC^{GluR1}* ablated discs bearing the fluorescent caspase reporter *UAS-GC3Ai* at 18 hr (A), 24 hr (B), 36 hr (C), and 48 hr (D). (E) A *DC^{GluR1}*>>*GC3Ai* ablated disc at 64 hr, with GC3Ai-positive cells present at the distal pouch after the nubbin marker is reestablished (arrowhead). (F) An area quantification of the GC3Ai signal in the whole pouch at 18 hr (n = 10), 24 hr (n = 3), 36 hr (n = 7), and 48 hr (n = 7), demonstrating that GC3Ai signal area increases over time. Data were analyzed with a one-way ANOVA followed by a multiple comparisons test. $P^* = 0.0308$, $P^{***} = 0.0001$, $P^{****} < 0.0001$. ns = not significant. Error bars represent the standard deviation. (G-G') A *DC^{GluR1}*>>*GC3Ai* ablated disc at 36 hr with GC3Ai-positive cells associated with the mitotic PH3 marker. Yellow arrowhead in (G') points to non-apoptotic GC3Ai-positive cells, while the white arrowhead points to GC3Ai labeled cells with an apoptotic morphology. (H-I) A *DC^{GluR1}* *LP>GC3Ai* ablated disc at 18 hr (H) and 36 hr (I). The open arrowhead in (H) points to the lack of GC3Ai labeling at the WE. The arrowhead in (I) highlights the presence of GC3Ai-labeled NiA cells at the wound edge. (J-J') A *DC^{hepCA}*>>*GC3Ai* ablated disc at 18 hr with apoptotic cells present throughout the disc proper. (K-K') A *DC^{hepCA}*>>*GC3Ai* ablated disc at 36 hr (K) and 64 hr (K'); all apoptotic cells appear to be extruded towards the basal surface of the epithelium. (L-L') A *DC^{GluR1}*>>*GC3Ai* ablated discs at 18 hr with NiA cells present in the disc proper. (M-M') A *DC^{GluR1}* *LP>GC3Ai* ablated disc at 36 hr, with columnar-shaped GC3Ai-positive cells in the disc proper (yellow arrowhead) alongside pyknotic, rounded GC3Ai-positive cells (red arrowhead), showing that there are a mix of morphologically apoptotic NiA and non-apoptotic NiCP cells in the disc proper. In each of the transverse sections (J', K', L', M'), Ba indicates the basal surface of the disc, while Ap indicates the apical surface of the disc. GC3Ai = green caspase-3 activity indicator, LP = lateral pouch. See Supplementary Genotypes file for exact genotypes.

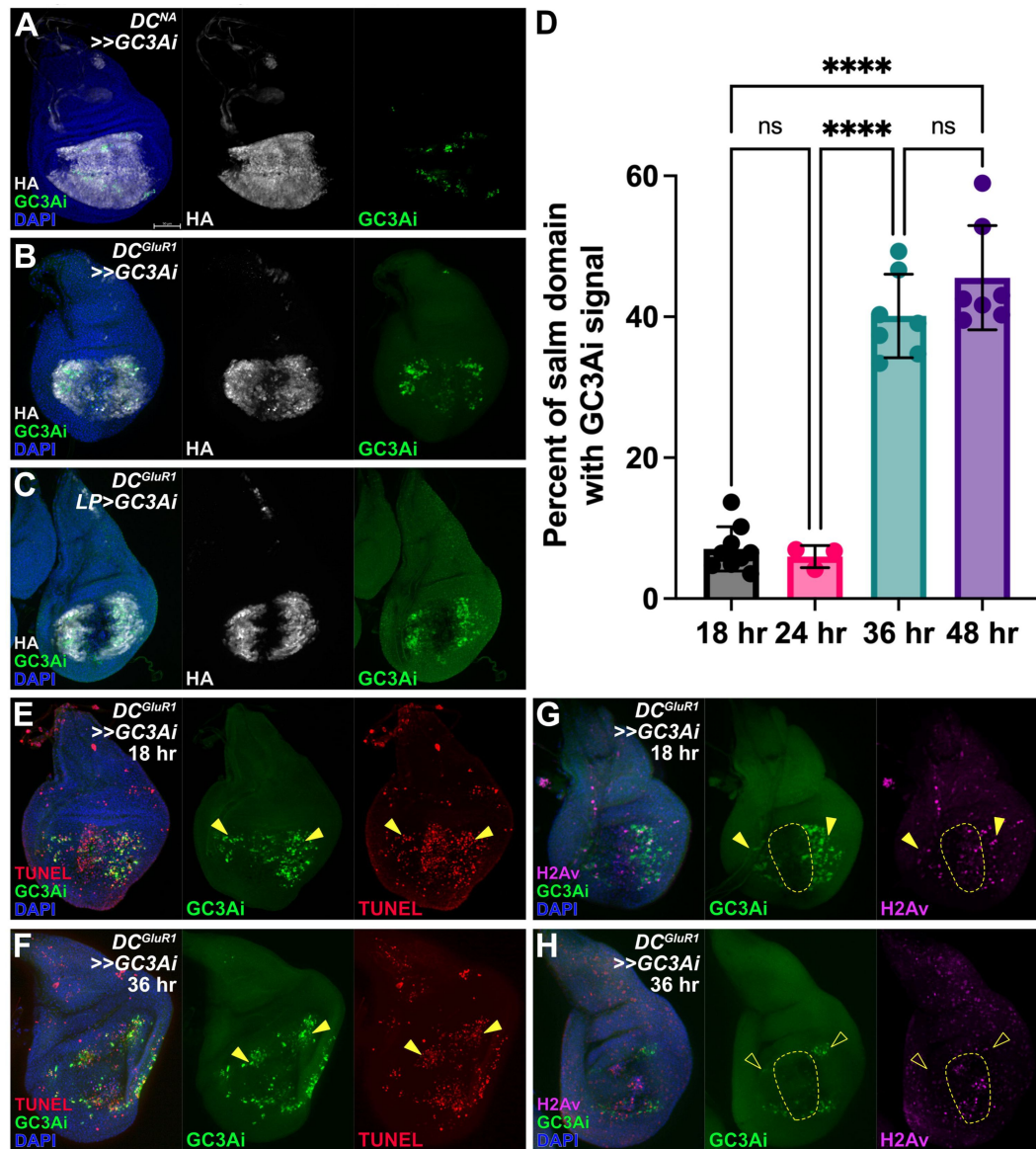


Figure 5 – figure supplement 1.

(A) A $DC^{NA}>GC3Ai$ control disc driving $UAS-GC3Ai$ in the pouch, HA labels all $GC3Ai$ -expressing cells, while the cells with activated $GC3Ai$ also have green fluorescence. (B) A $DC^{GluR1}>GC3Ai$ ablated disc showing $GC3Ai$ -positive NiA cells and, to a lesser extent, $GC3Ai$ -positive wound edge apoptotic cells. (C) A $DC^{GluR1} LP>GC3Ai$ ablated disc, with HA highlighting the restricted expression of $GC3Ai$ to the NiA area of the disc, thus excluding wound edge cells. (D) An area quantification of the $GC3Ai$ signal in the distal pouch at 18 hr ($n = 10$), 24 hr ($n = 3$), 36 hr ($n = 7$), and 48 hr ($n = 7$), demonstrating that $GC3Ai$ signal area increases over time. Data were analyzed with a one-way ANOVA followed by a multiple comparisons test. $P^{****} < 0.0001$. ns = not significant. Error bars represent the standard deviation. (E-F) DC^{GluR1} ablated discs at 18 hr (E) and 36 hr (F) with $GC3Ai/TUNEL$ double-positive NiA at both time points. (G-H) DC^{GluR1} ablated discs at 18 hr (G) and 36 hr (H) showing strong overlap between NiA and $\gamma H2Av$ at 18 hr (arrowheads in G) that diminishes by 36 hr (open arrowheads in H). $GC3Ai$ = green caspase-3 apoptosis indicator, LP = lateral pouch. See Supplementary Genotypes file for exact genotypes.

pyknotic nuclei (**Figure 5G** [↗](#), white arrowhead), while the majority appear to have a consistent and undisturbed cytoplasmic fluorescent label (**Figure 5G** [↗](#), yellow arrowhead). To test our hypothesis that these cells are not undergoing apoptosis, we performed a TUNEL assay to fluorescently label cells with double stranded DNA breaks in GC3Ai-expressing discs. We found that most GC3Ai-labeled cells are co-labeled by TUNEL in early (18 hr) and late (36 hr) stages of regeneration (**Figure 5 – figure supplement 1E** [↗](#) and **F** [↗](#), arrowheads). However, while TUNEL is associated with apoptosis, by itself it does not confirm that cells are dying (Grasl-Kraupp *et al.* 1995 [↗](#)). Therefore, we also examined levels of γ H2Av, a histone variant associated with DNA repair and inhibition of apoptosis following damaging stimuli such as irradiation (Madigan *et al.* 2002 [↗](#)). Interestingly, we found that GC3Ai-positive cells initially have high levels of γ H2Av at 18 hr of regeneration, including those at the WE and the NiA (**Figure 5 – figure supplement 1G** [↗](#), arrowheads), which later is lost from the NiA at 36 hr (**Figure 5 – figure supplement 1H** [↗](#), open arrowheads). This indicates that NiA are undergoing active DNA repair rather than apoptosis. Taken together, these data suggest that a majority of NiA may upregulate caspase activity, but rather than undergoing apoptosis, they repair cellular damage and persist in the tissue into late stages of regeneration where they promote proliferation. As such we are referring to this population of persistent and potentially non-apoptotic NiA as Necrosis-induced Caspase Positive (NiCP) cells.

NiCP cells have initiator caspase activity but sublethal effector caspase activity

While this characterization of caspase activation and discovery of NiCP cells do not change any of the conclusions we have made up until this point, it does raise the question as to why some cells (NiA) undergo apoptosis and are removed from the tissue in response to necrosis, while others (NiCP) seem to persist despite caspase activity. The ability for cells to survive caspase activity is not surprising, as many non-apoptotic roles for caspases have been documented (Su 2020 [↗](#)), and includes promoting proliferation. As mentioned, the initiator caspase Dronc functions in a non-apoptotic role to activate Duox and thus promote proliferation during AiP in damaged wing discs (Fogarty *et al.* 2016 [↗](#)). Therefore, we wondered whether the difference between NiA and NiCP might arise from the level or activity of caspases within these cells. The GC3Ai reporter indicates activity of Dcp-1 and Drice (Schott *et al.* 2017 [↗](#)) while the anti-cleaved-Dcp-1 antibody is thought to also detect Drice (Li *et al.* 2019 [↗](#)). Thus, these tools exclusively detect effector caspases. As such, to gain a better understanding of caspase activity in NiCP we used two lineage tracing tools to study the activity of initiator versus effector caspases. The initiator caspase-sensitive Drice-Based Sensor (*DBS-GFP*) consists of a modified version of the effector caspase Drice that is 1) a membrane-bound via mCD8, and 2) linked to a nuclear GFP (Baena-Lopez *et al.* 2018 [↗](#)). Upon cleavage by Dronc the GFP is released and translocates to the nucleus, providing a readout for Dronc activity. (Baena-Lopez *et al.* 2018 [↗](#)). The second tool, CasExpress, is also membrane-bound via mCD8 and attached to GAL4 through a short linker peptide that contains the Drice/Dcp-1 cleavage site DQVD, thus providing a readout for the activity of both effector caspases Drice and Dcp-1 (Ding *et al.* 2016 [↗](#)). This tool is highly sensitive and is capable of detecting non-apoptotic caspase activity that occurs during development. However, since GAL4 activity can be modulated by temperature, the sensitivity of CasExpress can be altered by a combination of GAL80^{ts} and temperature to suppress this background signal and optimize damage-induced caspase labeling (Colon Plaza and Su 2024 [↗](#)). Our results show a strong overlap of *DBS-GFP* with anti-cDcp-1 in the LP of *DC^{GluRI}* ablated discs (**Figure 6A** [↗](#) and A'), indicating that these cells have robust Dronc activity, as do the cells at the WE (**Figure 6A** [↗](#) and A'). However, when we used CasExpress to examine effector caspase activity with a protocol that eliminates developmental caspase activity (Colon Plaza and Su 2024 [↗](#)), we noted that only cells at the WE were labelled (**Figure 6B** [↗](#), B' and B'', arrowhead), most of which had pyknotic nuclei showing they are actively undergoing apoptosis, while few cells in the LP were labelled (**Figure 6B** [↗](#)). We hypothesized that the level of effector caspase activity might be high enough to be detected by GC3Ai and the cDcp-1 antibody, but not by CasExpress. To test this idea, we further attempted to detect NiCP with CasExpress by

combining it with G-TRACE (Evans *et al.* 2009 [↗](#)), which should lineage trace cells that have effector caspase activity, permanently labeling them with GFP at the start of ablation. Again, we found that only cells of the WE are labelled during regeneration (**Figure 6B** [↗](#), arrowhead), with minimal labelling of cells in the LP (**Figure 6B** [↗](#), open arrowhead), even after 36 hr of regeneration when the NiCP-induced uptick in proliferation occurs (**Figure 6** [↗](#) – figure supplement 1A). By comparison, performing these same experiments using *DC^{hepCA}* to induce extensive apoptotic cell death leads to a significant proportion of cDcp-1-positive cells being labelled by CasExpress under both normal and lineage tracing conditions (**Figure 6C** [↗](#), arrowhead), suggesting that in the context of necrosis there is not enough effector caspase activity to label NiCP using these methods. Indeed, if the CasExpress experiment is performed in the absence of the *GAL80^{TS}* that suppresses the background developmental caspase signal, the NiCP cells now become labeled by GFP (**Figure 6D** [↗](#)), indicating that effector caspases are indeed present in these cells, but at low enough levels to avoid death. Evidence for the existence of a cellular execution threshold of caspase activity in cells of the wing disc, which must be reached to induce apoptosis, has previously been documented (Florentin and Arama 2012 [↗](#)). This is further supported by the observation that the GFP label becomes more apparent later in regeneration (**Figure 6E** [↗](#)), confirming that cells with effector caspase activity persist rather than die. Thus, it appears that in response to necrosis, the cells of the LP activate the initiator caspase Dronc and the effector caspase(s) Drice/cDcp-1 (to an extent) but fail to undergo programmed cell death. Instead, these cells persist in the disc late into regeneration where they stimulate regenerative proliferation.

Dronc in NiCP cells promote proliferation independent of AiP

While NiCP cells have both initiator (Dronc) and effector caspase (Drice/cDcp-1) activity, it appears that the level or function of effector caspases is insufficient to cause apoptosis, and is also inconsequential for promoting regeneration – indeed, blocking Drice/cDcp-1 activity with P35 does not affect the localized regenerative proliferation observed at 36 hr (**Figure 4L and M** [↗](#)), or the overall ability to regenerate (Klemm *et al.* 2021 [↗](#)). This is in contrast with the ability of effector caspases to drive proliferation in the eye disc (Fan and Bergmann 2008 [↗](#)). However, blocking the apoptotic pathway upstream of Dronc using *mir(RHG)* eliminates NiA/NiCP (**Figure 7A** [↗](#), C and K, (Klemm *et al.* 2021 [↗](#)), blocks the increase in proliferation (**Figure 7B** [↗](#), D and L), and limits regeneration (Klemm *et al.* 2021 [↗](#)). These observations demonstrate that proliferation induced by NiCP must depend on factors downstream of *rpr/hid/grim* and upstream of the effector caspases *Drice/cDcp-1*, thus potentially pointing to a role for *Dronc*. To test this, we ablated discs with *DC^{GluR1}* while reducing the activity of Dronc using a null allele (*Dronc^{I29/+}*, (Xu *et al.* 2005 [↗](#))). With this genetic background there is a significant reduction in regenerative proliferation at 36 hr (**Figure 7F and L** [↗](#)) and regeneration is limited, shown by both adult wing scores (**Figure 7M** [↗](#)) and size (**Figure 7** [↗](#) – figure supplement 1A). These data demonstrate an essential role for Dronc in NiCP to promote proliferation and subsequent regeneration of the disc following necrosis. We noted that this mutant does not strongly affect the number of NiCP assayed by cDcp-1 (**Figure 7E and K** [↗](#)), likely because this allele does not completely suppress apoptosis unless homozygous (Xu *et al.* 2005 [↗](#)), which is precluded by the genetics of this experiment. As an alternative, we interfered with Dronc function by expressing a dominant negative form of *Dronc* that contains only the caspase recruitment domain (CARD), here referred to as *Dronc^{DN}*, which blocks activation of Dcp-1/Drice and apoptosis (Meier *et al.* 2000 [↗](#)). *Dronc^{DN}* reduces NiCP number, but to a lesser degree than *mi(RHG)* (**Figure 7G and K** [↗](#)) and does not affect proliferation (**Figure 7H and L** [↗](#)), suggesting that the CARD domain is dispensable for NiCP-induced proliferation.

Finally, we wondered how this might relate to the previously documented role of Dronc in promoting proliferation following apoptotic cell death during AiP (Fogarty and Bergmann 2017 [↗](#)). AiP depends on both JNK and ROS (Fogarty *et al.* 2016 [↗](#)), which we have shown are only present at the WE and are not associated with NiA/NiCP (**Figure 1** [↗](#) – figure supplement 1B, 4G, H and I). Thus, it is possible that Dronc's function in response to necrosis occurs via a distinct mechanism.

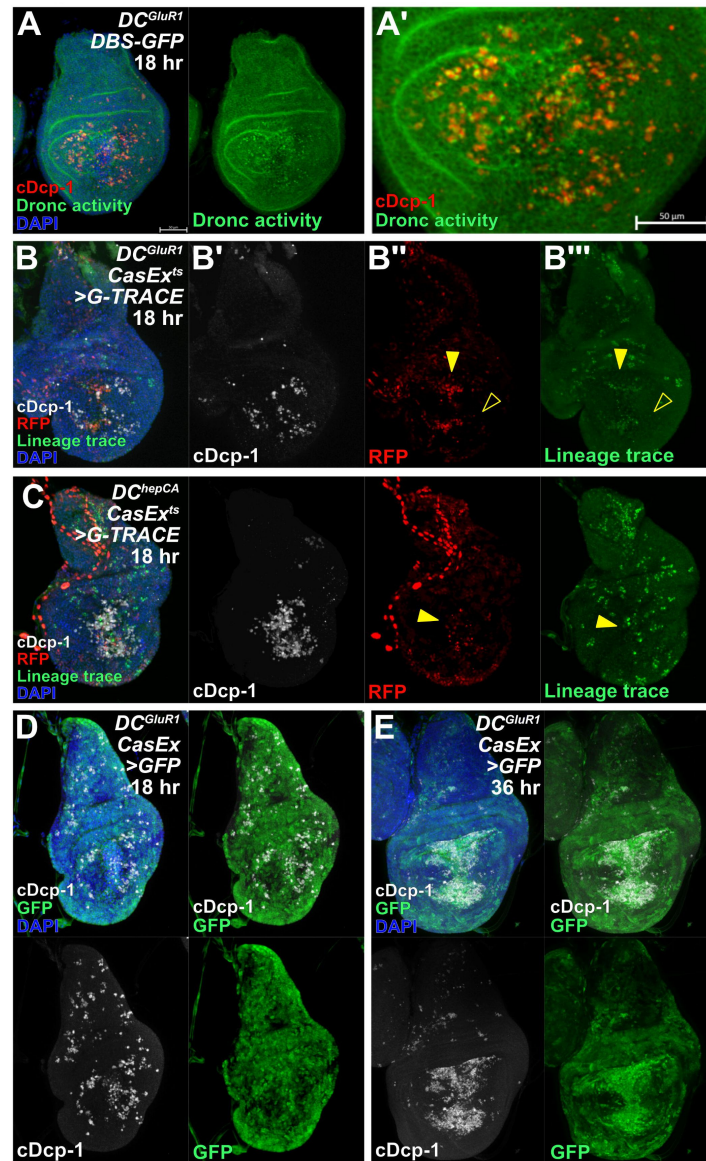


Figure 6

- NiCP have sublethal levels of activated executioner caspase.

(A-A') A *DC^{GluR1} DBS-GFP* ablated disc at 18 hr showing strong overlap between cDcp-1 and DBS-GFP-positive NiA/NiCP. (B-B''') *DC^{GluR1} CasEx^{ts} >G-TRACE* ablated discs at 18 hr with RFP labeled wound edge cells, (arrowhead), but no RFP or lineage trace-positive labeling is observed in NiA/NiCP (open arrowheads). (C) A *DC^{hepCA} CasEx^{ts} >G-TRACE* ablated disc at 18 hr showing a high level of overlap between apoptotic cells and both RFP and the lineage trace (arrowhead). (D-E) *DC^{GluR1} CasEx >GFP* ablated discs at 18 hr (D) and 36 hr (E) showing that NiA/NiCP are labeled by CasExpress in a sensitized background (when *tubGAL80^{ts}* is omitted). DBS = Drice-based sensor, CasEx = CasExpress. See Supplementary Genotypes file for exact genotypes.

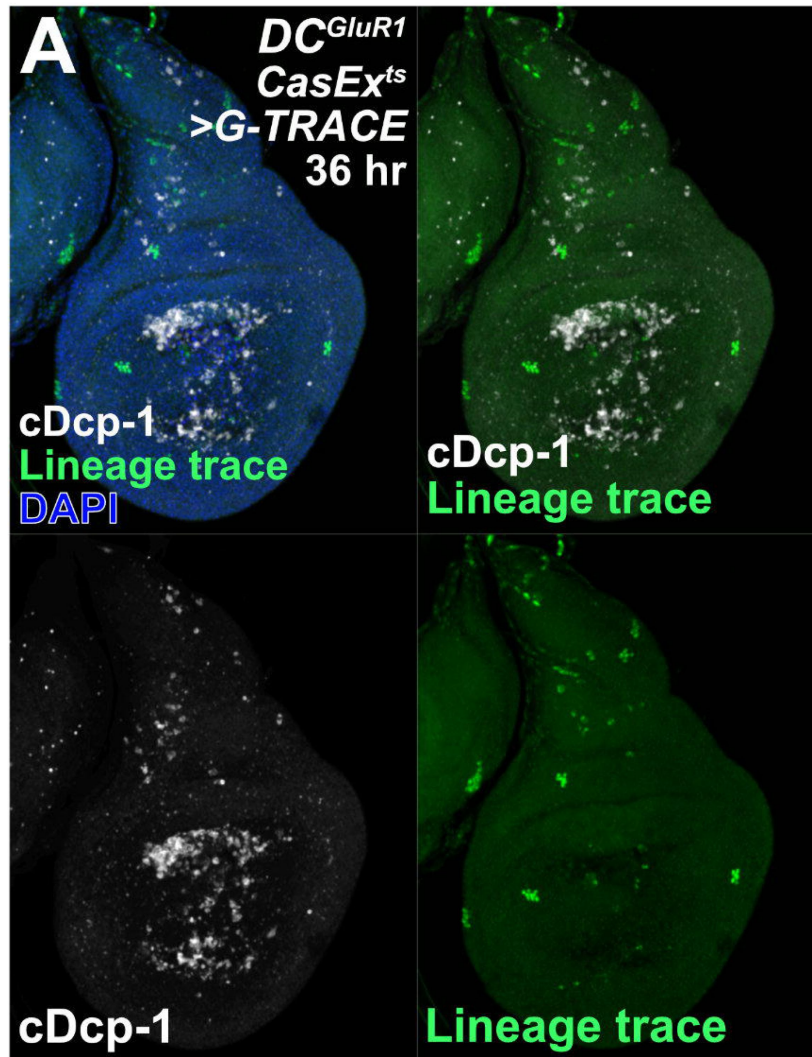


Figure 6 – figure supplement 1.

(A) A *DC^{GluR1} CasEx^{ts} >G-TRACE* ablated disc at 36 hr with no lineage-positive NiA/NiCP, despite the presence of clonal patches of cells traced by developmental caspase activity. See Supplementary Genotypes file for exact genotypes.

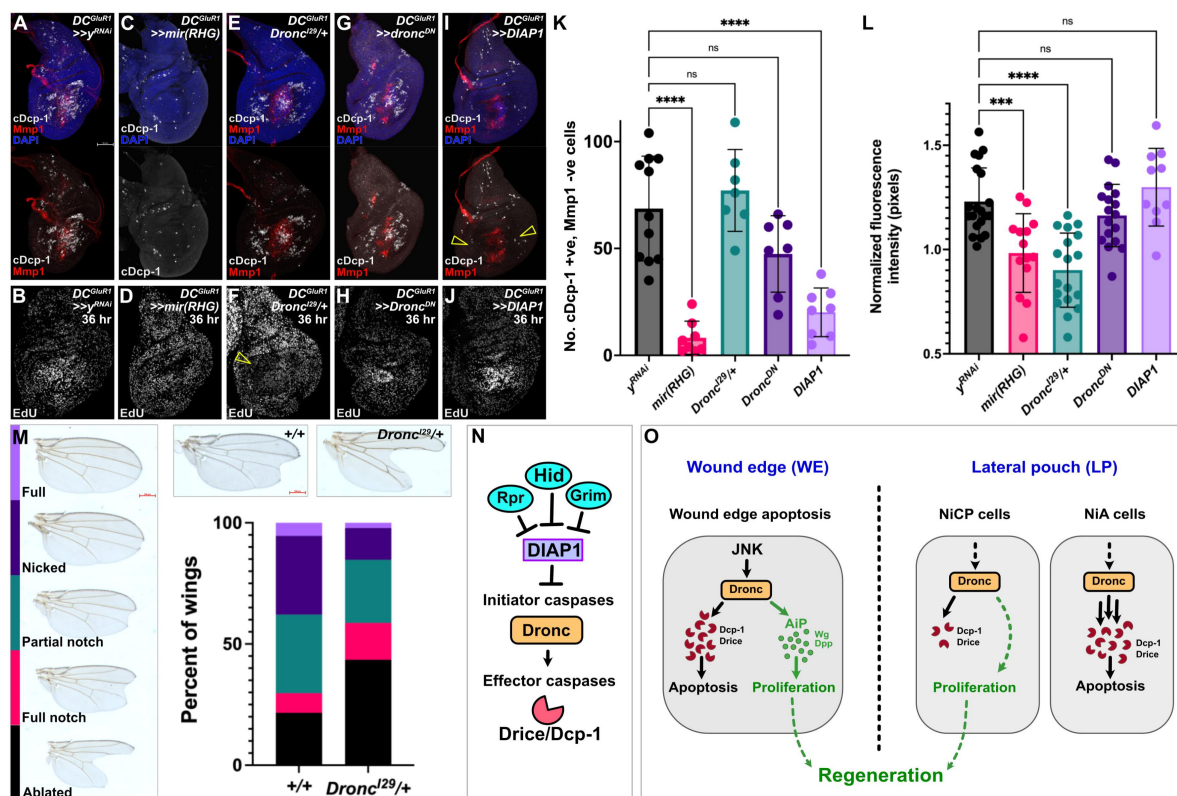


Figure 7

- Dronc activity in NiCP promotes regeneration.

(A-J) Representative *DC^{GluR1}* ablated discs at 18 hr (A, C, E, G and I) with cDcp-1 and Mmp1 labeling, and at 36 hr (B, D, F, H and J) with EdU labeling, bearing *UAS-y^{RNAi}* (A-B), *UAS-mir(RHG)* (C-D), *Dronc^{I29/+}* (E-F), *UAS-Dronc^{DN}* (G-H), and *UAS-DIAP1* (I-J). Open arrowhead in (F) indicates the strong reduction in EdU labeling. Open arrowheads in (I) show a strong suppression of NiA/NiCP as detected by cDcp-1. (K) A quantification of NiA/NiCP number (cDcp-1+ve, Mmp1-ve cells) in *UAS-y^{RNAi}* (n = 11), *UAS-mir(RHG)* (n = 8), *Dronc^{I29/+}* (n = 10), *UAS-UAS-Dronc^{DN}* (n = 7), and *UAS-DIAP1* (n = 8), ns, not significant, P**** < 0.0001. Data were analyzed with a one-way ANOVA followed by a multiple comparisons test. Error bars represent the standard deviation. (L) A quantification of EdU fluorescence intensity with *UAS-y^{RNAi}* (n = 20), *UAS-mir(RHG)* (n = 14), *Dronc^{I29/+}* (n = 18) *UAS-UAS-Dronc^{DN}* (n = 17) and *DIAP1* (n = 10), ns, not significant, P*** = 0.0004, P**** < 0.0001. Data were analyzed with a one-way ANOVA followed by a multiple comparisons test. Error bars represent the standard deviation. (M) Regeneration scoring of *DC^{GluR1}* ablated flies in a +/+ versus a *Dronc^{I29/+}* background. Note the increase in frequency of full notched and ablated wings in *Dronc^{I29/+}* relative to the +/+ treatment. (N-O) Schematics depicting the apoptotic pathway in *Drosophila* (N) and the three different caspase-positive cell types that occur following necrotic ablation (O). Following necrosis, JNK signaling at the wound edge (WE) promotes Dronc activation, which mediates WE apoptosis and proliferation through the AIP feedback loop (left). In the lateral pouch (LP), DAMP-like signal(s) leads to the formation of necrosis-induced caspase positive (NiCP) cells that utilize Dronc to promote proliferation and subsequent regeneration independent of both JNK and AIP signaling, while necrosis-induced apoptosis (NiA) undergo apoptotic death resulting from high levels of dronc, cDcp-1 and Drice activity (right). Both WE apoptosis and NiCP act to promote regeneration through separate mechanisms relying on Dronc-mediated regenerative proliferation. See Supplementary Genotypes file for exact genotypes.

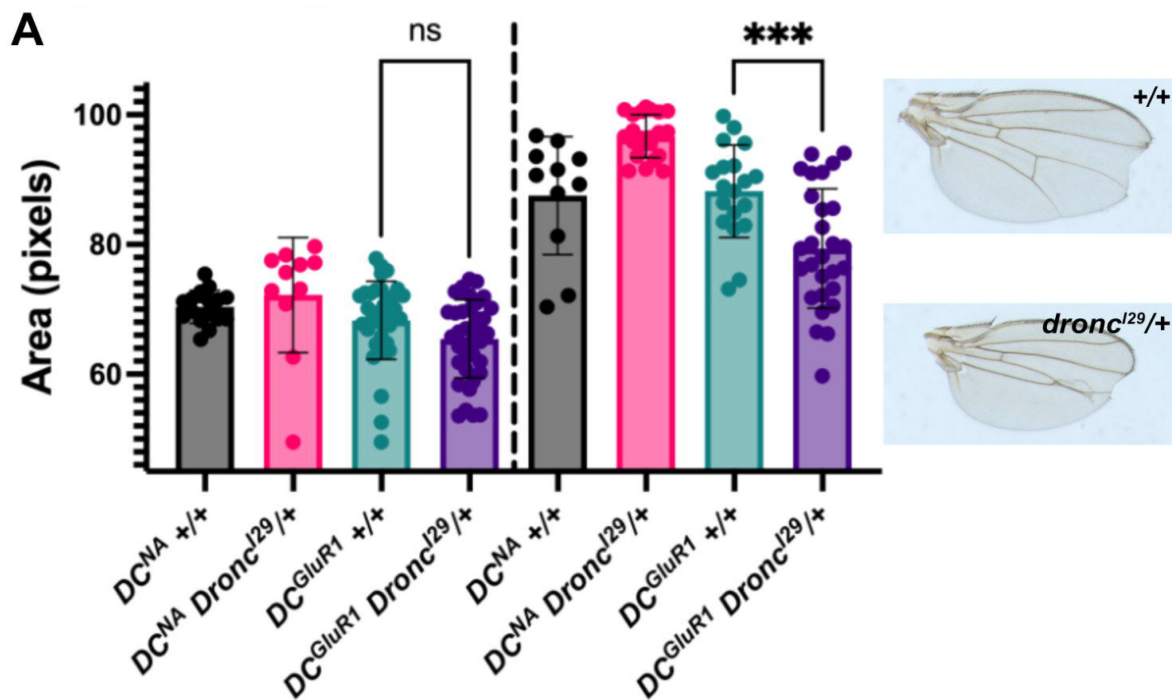


Figure 7 – figure supplement 1.

(A) A quantification of regenerated wing size in an unablated control (DC^{NA}) wildtype (+/+), males = 17, females = 11) and $Dronc^{I29}/+$ (males = 11, females = 22) background, and following DC^{GluR1} ablation in a +/+ (n = 29 males, n = 19 females) and $Dronc^{I29}/+$ (n = 24 males, n = 28 females) background, ns = not significant, $P^{***} = 0.0001$, data were analyzed with a one-way ANOVA followed by a multiple comparisons test. Error bars represent the standard deviation. Representative images for regenerated +/+ (top) and $Dronc^{I29}/+$ (bottom) wings are shown to the right. See Supplementary Genotypes file for exact genotypes.

Importantly, the activity of Dronc in both apoptosis and in AiP is influenced by the Dronc's upstream regulator, DIAP1 (Meier *et al.* 2000 [↗](#)). DIAP1 modifies Dronc's CARD domain through K78 mono-ubiquitylation, which blocks both apoptosis (Kamber Kaya *et al.* 2017 [↗](#)), and AiP (Fan and Bergmann 2008 [↗](#), Fogarty and Bergmann 2017 [↗](#)). Thus, we expressed *DIAP1* (*UAS-DIAP1*) following ablation with *DC^{GluRI}* and found that it strongly suppresses the number of cDcp-1-positive cells at the WE and the NiA/NiCP in the LP (**Figure 7I and K** [↗](#)), but strikingly has no effect on regenerative proliferation at 36 hr (**Figure 7J and L** [↗](#)). These results demonstrate a key role for the initiator caspase *Dronc* in promoting regenerative proliferation following necrosis, which is not affected by DIAP1, and therefore is likely separate from its role in apoptosis and the AiP mechanism.

Taken together, our data suggest a model in which necrosis leads to the establishment of distinct cell populations important for regeneration (**Figure 7N and O** [↗](#)). After injury, cells at the immediate WE undergo JNK-mediated apoptosis and contribute to proliferation via the established AiP mechanism (**Figure 7O** [↗](#), left), while cells at a distance from the injury in the LP activate Dronc via an unknown DAMP-like signal(s) that occurs independent of JNK. Some of these cells go on to activate effector caspases at levels high enough to result in apoptosis (NiA, **Figure 7O** [↗](#), right), while others activate these caspases at a low enough level to be detectable but insufficient to induce death (NiCP, **Figure 7O** [↗](#) right). Instead, these cells persist in the tissue late into the repair process, where they promote proliferation via a novel non-apoptotic and AiP-independent function of the initiator caspase Dronc.

Discussion

An important and often overlooked factor when studying regeneration is the type of injury – and consequently the type of cell death it causes – that can significantly impact repair processes. The existence of conserved signaling events that promote recovery has been well established in the context of apoptosis (Fogarty and Bergmann 2017 [↗](#), Perez-Garijo 2018 [↗](#)), while the importance of such events in necrosis are less well established. Here, we have investigated the genetic events that occur in the aftermath of necrosis and how they influence the ability of a tissue to recover and regenerate. We previously showed that wing discs can regenerate effectively in the face of necrotic cell death (Klemm *et al.* 2021 [↗](#)), triggering local JNK-dependent AiP at the WE, which is likely in response to tissue disruption (La Marca and Richardson 2020 [↗](#)), while also inducing caspase activity in cells at a distance from the wound. This induction is independent of JNK signaling and, alongside having effector caspase activity, these cells can be marked by TUNEL and can be blocked by inhibiting the apoptotic pathway (Klemm *et al.* 2021 [↗](#)). As such, we determined that these cells are undergoing PCD and named this phenomenon Necrosis-induced Apoptosis (NiA). We also showed that both NiA and AiP at the wound contribute to regeneration. Our current work further characterizes the NiA phenomenon, and we have now shown that cells undergoing NiA actually comprise two populations with separate behaviors. Upon necrosis, cells of the LP appear to activate effector caspases, indicated by cDcp-1 antibody staining and the transgenic reporter GC3Ai. While a proportion of these cells develop apoptotic morphology, round up and are cleared over time as part of NiA, a large number of cells appear able to persist despite the presence of caspases, where they promote regenerative proliferation dependent on the initiator caspase Dronc (**Figure 7N** [↗](#)). To reflect these new findings, we have called these Necrosis-induced Caspase Positive (NiCP) cells. While NiCP appear to promote proliferation, it remains unclear whether they do so through cell autonomous or non-autonomous effects. We were able to determine that NiCP do not utilize AiP as a potential mechanism, as these events occur in the absence of JNK, and exhibit none of the established hallmarks of AiP, including the presence and requirement for ROS, the ability to be blocked by DIAP1, or the production of mitogens such as Wg and Dpp. However, it is possible that other signaling factors may be secreted by these NiCP cells. Alternatively, the observed non-apoptotic caspase activity in NiCP cells may interact with

intracellular factors to promote cell-autonomous proliferation. In either case, our results have identified an important role for Dronc in promoting regeneration in response to necrotic cell death.

The signal from necrotic cells that leads to NiA/NiCP remains unknown

Although we have shown that necrosis leads to NiCP cells and NiA at a distance from the site of injury, the signal that leads to these events upon wounding is still unknown. Cells undergoing necrosis release DAMPs, a category of molecules that includes both common cellular contents as well as specific proteins, both of which can produce downstream responses like inflammation and the activation of effectors that promote a healing response (Gordon *et al.* 2018 [↗](#)). DAMPs of both categories have been demonstrated in *Drosophila*; in apoptotic-deficient larvae, circulating DAMPs in the hemolymph can constitutively activate immune signaling in the fat body (Nishida *et al.* 2024 [↗](#)), while specific factors such as α -actinin (Gordon *et al.* 2018 [↗](#)), and HMGB1 (Nishida *et al.* 2024 [↗](#)) are known to possess DAMP activity. Recent work describing an *in vivo* sensor for HMGB1 demonstrates the release of this DAMP from wing disc cells in response to a necrotic stimulus similar to that used here (Nishida *et al.* 2024 [↗](#)), making this an important candidate to test for a potential role in producing NiCP. However, it is equally possible that the causative DAMP(s) is/are one or more of the common fundamental cellular components released upon cell lysis, which would be more challenging to manipulate and test. Considering the potential diversity and variable nature of DAMPs, it may be more feasible to instead identify the downstream PRRs that are required to interpret this unknown signal. Several groups of genes can act as PRRs, most notably the Toll-like receptor (TLR) family, various members of which can respond to DAMPs (Ming *et al.* 2014 [↗](#), Gong *et al.* 2020 [↗](#)). Nine TLR genes have been identified in *Drosophila*, which have assorted roles in development and innate immunity (Anthoney *et al.* 2018 [↗](#)), and have been implicated in DAMP sensing (Ming *et al.* 2014 [↗](#)). Thus, it would be valuable to test the requirement for TLRs, alongside other suspected PRRs such as scavenger receptors (Cao 2016 [↗](#), Roh and Sohn 2018 [↗](#), Gong *et al.* 2020 [↗](#)) for their requirement to cause NiCP.

An additional approach to elucidating how these cells are generated is by leveraging our finding that both JAK/STAT and WNT signaling seem to block NiCP/NiA in the disc. JAK/STAT signaling promotes survival of cells in response to stress by repressing JNK signaling, thus minimizing JNK-mediated apoptosis (La Fortezza *et al.* 2016 [↗](#)). Although NiCP appear to avoid damage-induced JAK/STAT upregulation in the pouch, it is unlikely they are regulated by this mechanism, since NiCP cells occur independent of JNK activity (Klemm *et al.* 2021 [↗](#)). However, within the hinge, which is completely devoid of NiCP, JAK/STAT protects cells from apoptosis potentially by upregulating *DIAP1* (Verghese and Su 2016 [↗](#)) and *zfh2* (Verghese and Su 2018 [↗](#)), while Wg represses the transcription of *rpr*, (Verghese and Su 2016 [↗](#)). Both signaling pathways are required autonomously in hinge cells for their ability to replace and regenerate ablated pouch tissue (Verghese and Su 2016 [↗](#), Ledru *et al.* 2022 [↗](#)). Our results show that DIAP1 can prevent cDcp-1 activation in NiCP in the pouch, but not the corresponding proliferation that they produce, suggesting one reason that NiCP is not seen in the hinge is possibly due to the high DIAP1 threshold that results from developmental JAK/STAT activity, which blocks Dronc's ability to activate effector caspases in this region. Thus, it remains to be seen whether lowering DIAP1 levels in the hinge, or manipulating Wg to allow *rpr* expression alongside changes in JAK/STAT, could allow NiCP to form in the hinge, and whether this might lead to regenerative proliferation in this region. Finally, the observation that both NiCP cells and their associated proliferative effects appear to occur independent of JNK signaling, which is normally central in models of stress and damage (Pinal *et al.* 2019 [↗](#)), could help to identify the upstream signaling that leads to NiCP following necrosis. Together, these approaches could provide essential information as to the underlying genetic and cellular events that connect the lysis of cells during necrosis to the formation of NiCP cells required for regeneration.

The persistence of NiCP cells versus their elimination via NiA

Our work shows that cells in the LP can either persist as NiCP and contribute to regenerative proliferation, or progress to apoptosis as part of NiA, but how this decision is made is unknown. During apoptosis in the wing disc, a threshold level of effector caspases must be reached for the cell to complete PCD (Florentin and Arama 2012 [↗](#)). Thus, we hypothesize that DAMP signals from necrotic cells may result in inconstant levels of effector caspase activity in cells of the LP – some with high caspases that advance to apoptosis (NiA), recognized by the changes in morphology and position in the disc characteristic of PCD, which are ultimately cleared, and others (the NiCP cells) that have low enough caspase activity levels to survive, but this activity can still be detected by sensitized reagents. This is supported by our observations made using effector caspase-based lineage tracing using CasExpress (Ding *et al.* 2016 [↗](#)), in which caspases can only be detected in NiCP cells by lowering the detection threshold. Alternatively, since CasExpress is a membrane-based reporter (Ding *et al.* 2016 [↗](#)), it is also possible that effector caspases within NiCP cells have a different subcellular localization that reflects a non-apoptotic function, thus preventing reporter activation, rather than a difference in expression level or activity. The idea that caspases have different functions, both apoptotic and non-apoptotic, based on localization is well established, such as the targeting of specific cleavage substrates in distinct subcellular compartments (Brown-Suedel and Bouchier-Hayes 2020 [↗](#)), or the trafficking of Dronc to the membrane to promote ROS (Amcheslavsky *et al.* 2018 [↗](#)). Whatever the mechanism is that distinguishes between elimination via NiA or persistence as NiCP, the proportion of LP cells that participate in each remains difficult to assay due to the dynamic nature of cell death (Nano and Montell 2024 [↗](#)) and overall variability of the NiCP/NiA phenotype. Anecdotal evidence from our experiments examining GC3Ai-labelled cells over time suggests the majority of cells in fact do not undergo apoptosis and are therefore NiCP. Importantly, however, while our experiments demonstrate that GC3Ai labeled NiCP increase over time, it is unclear whether this increase in signal abundance is due to cells with newly activated caspases that cleave and activate GC3Ai, or if GC3Ai labeled NiCP are instead proliferating. The use of other tools with potentially different sensitivities to effector caspases, such as Apoliner (Bardet *et al.* 2008 [↗](#)), or CD8-PARP-Venus (Florentin and Arama 2012 [↗](#)), may shed further light on this issue. Nevertheless, it remains to be seen how a cell becomes NiCP or undergoes NiA and dies, and whether distinct levels of caspases (initiator or effector) their localization, or a different attribute is responsible.

How does NiCP contribute to regeneration?

One of the most important questions that future work must address is how the phenomena we have identified lead to regeneration following necrosis. Our experiments suggest that different caspase-positive populations of cells may contribute to regenerative proliferation: Firstly, the smaller number of apoptotic cells at the WE generated by epithelial disruption likely contribute by the established process of JNK-dependent AiP (Fogarty and Bergmann 2017 [↗](#)). Secondly, the caspase-surviving NiCP cells in the LP that contribute via the initiator caspase Dronc. Crucially, this non-apoptotic function of Dronc cannot be inhibited by the expression of DIAP1, which has previously been shown to block both the apoptotic and non-apoptotic AiP functions of Dronc (Kamber Kaya *et al.* 2017 [↗](#)). Thus, our results suggest that Dronc acts via a different mechanism to induce growth in response to necrosis. The current understanding of how Dronc functions at the molecular level may provide valuable clues as to how. DIAP1 suppresses Dronc through its E3 ubiquitin ligase activity, which mono-ubiquitylates the caspase recruitment (CARD) domain of Dronc to suppress both apoptotic and non-apoptotic functions related to AiP (Kamber Kaya *et al.* 2017 [↗](#)). Here, we find that the ectopic expression of *DIAP1* or just the CARD-containing pro-domain of Dronc (*Dronc^{DN}*) cannot block regenerative proliferation, while by contrast heterozygosity for the *Dronc^{J29}* null allele that can inhibit AiP (Kamber Kaya *et al.* 2017 [↗](#)) is sufficient to block NiCP-mediated proliferation. Thus, regulation of Dronc activity in the context of necrosis may not rely on the modification of its CARD domain. The documented functions of Dronc have also shown a requirement for its catalytic domain; mutations in this domain can block the

activation of Drice and AiP-induced overgrowth (Fan *et al.* 2014 [↗](#)). Thus, the inability of DIAP1 to suppress NiCP-mediated proliferation may be due to the catalytic activity of Dronc. Importantly, although the CARD and catalytic domains are important points of regulation for Dronc activity, it has also been shown that damage-specific context cues are also vital. For example, the ectopic expression of *Dronc*^{K78R}, a mutant that renders it irrepressible by DIAP1, might be expected to induce significant apoptotic cell death. However, this is not the case unless its structural binding partner, the APAF1 ortholog encoded by *Dark*, is expressed alongside (Shapiro *et al.* 2008 [↗](#)), demonstrating the importance of stoichiometry between Dronc and Dark for the apoptotic function of Dronc. As such, further investigation into Dronc's functional domains and context-dependent interactions with respect to its CARD and catalytic domains as well as its binding partners, including DIAP1 and Dark, will be necessary to understand how Dronc is involved in promoting regenerative proliferation in response to necrotic injury.

NiCP as a general mechanism to promote regeneration

Although having initially been characterized for their central role in apoptosis, many non-apoptotic functions of caspases have since been discovered, revealing them to be dynamic regulators of diverse process including cell fate specification, cellular remodeling, tissue growth, development, metabolism and others (Shinoda *et al.* 2019 [↗](#), Wang and Baker 2019 [↗](#), Su 2020 [↗](#)). Studies of caspase signaling during regeneration have revealed essential aspects of non-apoptotic caspase activity, such as initiator and effector-dependent models of AiP that contribute to repair (Ryoo and Bergmann 2012 [↗](#), Fogarty and Bergmann 2017 [↗](#)). The contrasts in caspase functions that we have observed between apoptotic and necrotic damage, despite ultimately resulting in comparable levels of regeneration in an organ (Klemm *et al.* 2021 [↗](#)), underscores the nuance that exists in damage signaling between different injury contexts. It is clear that caspase activity in response to injury as a mechanism to promote regeneration is a highly conserved process that occurs in many organisms, regardless of tissue identify or type of damage incurred (Bergmann and Steller 2010 [↗](#), Vríz *et al.* 2014 [↗](#), Fuchs and Steller 2015 [↗](#), Perez-Garijo and Steller 2015 [↗](#), Fogarty and Bergmann 2017 [↗](#), Perez-Garijo 2018 [↗](#)). Our findings reinforce the idea that much remains to be understood about the role of caspases in tissue repair, and that the nature of an injury – and therefore the type of cell death involved – is an important aspect that must also be considered.

Materials and Methods

Drosophila stocks

Flies were cultured in conventional dextrose fly media at 25°C with 12h light–dark cycles. The recipe for dextrose media contains 9.3 g agar, 32 g yeast, 61 g cornmeal, 129 g dextrose, and 14 g tegosept in 1 L distilled water. Genotypes for each figure panel are listed in the Supplementary Genotypes file. Fly lines used as ablation stocks are as follows: hs-FLP; hs-p65; salm-LexADBD, DVE>>GAL4 (DC^{NA}), hs-FLP; hs-p65; salm-LexADBD/ TM6C, sb (DC^{NA} no GAL4), hs-FLP; lexAop-GluR1^{LC}, hs-p65/ CyO; salm-LexADBD, DVE>>GAL4/ TM6B, Tb (DC^{GluR1}), hs-FLP; lexAop-GluR1^{LC}, hs-p65/CyO; salm-LexADBD/TM6C, sb (DC^{GluR1} no GAL4), hs-FLP;lexAOp-GluR1^{LC}/ CyO; salm-LexADBD, hh-GAL4/ TM6B, Tb (DC^{GluR1} hh-GAL4), hs-FLP;lexAop-hepCA, hs-p65/CyO; salm-DBD, DVE>>GAL4/ TM6B, Tb (DC^{hepCA}), UAS-GluR1; tubGAL80^{ts}, and tubGAL80^{ts}. The stock DR^{WNT}-GAL80 was used to limit *UAS*- transgenes to the lateral pouch (LP) where NiA occur (Klemm *et al.* 2021 [↗](#)), while *R85E08-GAL4* was used to drive *UAS*- expression at the wound edge (WE). The following stocks were obtained from Bloomington Drosophila Stock Center: w¹¹¹⁸ (BL#3605), UAS-y^{RNAi} (BL#64527), UAS-p35 (BL#5073), UAS-GC3Ai (II, BL#84346), UAS-GC3Ai (III, BL#84343), rn-GAL4 (BL#7405), hh-GAL4 (BL#600186), ptc-GAL4 (BL#2017), pnr-GAL4 (BL#), (BL#25758), nub-GAL4 (BL#25754), R73G07-GAL4(BL#39829), UAS-Zfh2^{RNAi} (BL#50643), UAS-wg^{RNAi} (BL#32994), UAS-Stat92E^{RNAi} (BL#35600), dpp-lacZ (BL#8412), wg-lacZ (BL#50763), spi-lacZ (BL#10462), UAS-p35 (II, BL#5072), UAS-p35 (III, BL#5073), AP-1-GFP (Chatterjee and Bohmann 2012 [↗](#)), act>>GAL4,

UAS-RFP (BL#30558), DBS-GFP (III, BL#83130), CasExpress (BL#65419), G-TRACE (III, BL#28281), tubGAL80^{ts} (II, BL#7019), tubGAL80^{ts} (III, BL#7017), 10xSTAT-GFP (BL#), UAS-dome^{RNAi} (BL#32860), UAS-hop48A (BL#), PCNA-GFP (BL#25749), UAS-Cat (BL#24621), UAS-Sod1 (BL#24754), UAS-Duox^{RNAi} (BL#32903), mol-lacZ (BL#12173), UAS-Dronc^{DN} (BL#58992), UAS-DIAP1 (BL#6657), and dronc^{I29}/TM3, Sb (BL#98453). UAS-mir(RHG) was gifted from the Hariharan lab at UC Berkeley. vgQE-lacZ was gifted from Tin Tin Su. UAS-GluR1^{LC} (Liu *et al.* 2013 [DOI](#)) was gifted from the Xie lab at Stowers Institute.

Ablation experiments

DUAL Control ablation with *DVE>>GAL4*

DUAL Control experiments were performed essentially as described in Harris *et al.* (2020) [DOI](#). Briefly, experimental crosses were cultured at 25°C and density controlled at 50 larvae per vial. Larvae were heat shocked on day 3.5 of development (84 hr after egg deposition (AED)) by placing vials in a 37°C water bath for 45 min, followed by a return to 25°C. Larvae were allowed to recover for 18 hr before being dissected, fixed and immunolabeled, unless otherwise indicated. UAS-*y^{RNAi}* and UAS-GFP were used as control lines for RNAi-based experiments. *w¹¹¹⁸* was used as a control for *dronc^{I29}* experiments. The *DVE>>GAL4* driver drives expression in the wing pouch, allowing for the regenerating wound edge cells and NiA cells to be targeted for interrogation (Klemm *et al.* 2021 [DOI](#)). The *DR^{WNT}-GAL80* transgene (Klemm *et al.* 2021 [DOI](#)) was included as necessary to restrict UAS- expression to the lateral pouch (LP), where NiA/NiCP occurs.

DUAL Control ablation without *DVE>>GAL4*

To restrict UAS- expression to wound edge apoptotic cells, a version of DUAL Control lacking the *DVE>>GAL4* (*DC^{GluR1} no GAL4*) was crossed to the *R85E08-GAL-4* driver. *DC^{GluR1} no GAL4* experiments were performed along the same parameters as *DC^{GluR1}* experiments. The *R85E08-GAL4* driver was used alongside *DC^{GluR1} no GAL4*, when necessary, to target UAS- expression to the wound edge (WE), where the JNK-dependent wound edge apoptotic cells occur.

DUAL Control ablation with *hh-GAL4*

DUAL Control flies bearing *hh-GAL4* (*DC^{GluR1} hh^{ts}*) were cultured at 18°C and density controlled at 50 larvae per vial. *tubGAL80^{ts}* was included to conditionally express UAS- based constructs after ablation. Larvae were heat-shocked on day 7 of development (168 hr AED) for 45 min at 37°C, followed by incubation at 30°C to inactivate *tubGAL80^{ts}* and permit UAS-based expression. Larvae were held at 30°C for 18 hr before being dissected, fixed, and immunolabeled.

GAL4/UAS ablation

GAL4/UAS-based ablation experiments were performed essentially as described in Smith-Bolton *et al.* (2009). Briefly, larvae bearing either *UAS-GluR1;tubGAL80^{ts}* were cultured at 18°C and density controlled at 50 larvae per vial. Larvae upshifted on day 7 of development (168 hr AED) for 20 hr at 30°C and were either immediately dissected (denoted as 0 hr) or were allowed to recover for 24 hr before being dissected, fixed, and imaged. *tubGAL80^{ts}* flies were used as a non-ablating control.

FLP/FRT ablation experiments

To generate clonal patches of *UAS-GluR1;UAS-RFP*-expressing cells, flies of the genotype *hs-FLP; AP-1-GFP; act>>GAL4, UAS-RFP/ T(2:3)SM6A, TM6B, Tb* were crossed to flies bearing *UAS-GluR1; tubGAL80^{ts}*. Larvae were cultured at 18°C and heat shocked in a 37°C water bath for 10 min at 82 hr AEL, returned to 18°C, and upshifted to 30°C for 20 hr at 168 h AEL, followed by dissection and immunostaining. *tubGAL80^{ts}* flies were used as a non-ablating control.

Regeneration scoring and wing measurements

Wings of adult flies from heat shocked larvae were scored and measured after genotype blinding by another researcher. Scoring was performed on anesthetized adults by binning into a regeneration scoring category (Harris *et al.* 2020 [DOI](#), Klemm *et al.* 2021 [DOI](#)). Wing measurements were performed by removing wings, mounting in Permount solution (Fisher Scientific) and imaged using a Zeiss Discovery.V8 microscope. Wing area was measured using the Fiji software. Male and female adults were measured separately to account for sex differences in wing size using a reproducible measuring protocol that excludes the variable hinge region of the wing (details of measuring protocol available on request). Statistics were performed using GraphPad Prism 10.0.

Immunohistochemistry

Larvae were dissected in 1 x PBS followed by a 20 min fix in 4 % paraformaldehyde in PBS (PFA). After 3 washes in 0.1 % PBST (1 x PBS + 0.1 % Triton-X), larvae were washed in 0.3% PBST and then blocked in 0.1 % PBST with 5 % normal goat serum (NGS) for 30 min. Primary staining was done overnight at 4°C, and secondary staining was done for 4 hr at room temperature. The following primary antibodies were obtained from the Developmental Studies Hybridoma Bank: mouse anti-Nubbin (1:25), mouse anti-Wg (1:100), mouse anti-Mmp1 C-terminus (1:100), mouse anti-Mmp1 catalytic domain (1:100), mouse anti-LacZ (1:100), mouse anti-discs large (1:50), mouse anti-yH2Av (1:100), and rat anti-DE-cadherin (1:100). Rabbit anti-cDcp-1 (1:1000), mouse anti-PH3 (1:500), and rabbit anti-HA (1:1000) were obtained from Cell Signaling Technologies. Rat anti-Zfh-2 was generously gifted by Chris Doe. Anti-rabbit 647, anti-rat 647, anti-mouse 555, and anti-mouse 488 secondary antibodies were obtained from Invitrogen and used at a 1:500 dilution. DAPI (1:1000) was used as a counterstain. Images were obtained on a Zeiss AxioImager.M2 with ApoTome and a Leica TCS SP8 LCSM (NIH SIG award 1 S10 OD023691-01) housed in the Regenerative Medicine Imaging Facility at Arizona State University. For each experiment at least 15 discs were analyzed prior to choosing a representative image. Images were processed using Affinity Photo.

EdU staining, TUNEL assay, and DHE staining

EdU

The Click-It EdU Alexa Fluor 555 Imaging Kit (Invitrogen C10338) was used to assay cell proliferation. Briefly, imaginal discs were dissected and labeled with 1 μ l EdU in 1 ml PBS for 20 min, fixed in 4 % PFA for 20 min and immunolabeled (as necessary), followed by a 30 min Click-It reaction that was performed as directed in the EdU manual.

TUNEL

The TUNEL assay was performed with the ApopTag Red In Situ Apoptosis Detection Kit (Millipore S7165). Dissected larvae were fixed in 4 % paraformaldehyde, followed by a 10 min wash in 75 ml equilibration buffer. Discs were then submerged in 55 μ l working strength TdT enzyme for 3 hr at 37°C. The reaction was stopped by adding 1ml stop/wash buffer and incubating for 10min at room temperature, followed by three washes in PBS. Immunolabeling was performed by incubating the tissue preps with 65 μ l of anti-digoxigenin rhodamine overnight at room temperature.

Dihydroethidium (DHE)

DHE labeling was performed by incubating freshly dissected wing imaginal discs in Schneider's Media with 1 μ l of 10 mM DHE reconstituted in 1 ml DMSO (for a working concentration of 10 μ M DHE) for 10 min, followed by three 5 min washes in PBS and immediately mounting and imaging.

Quantification and Statistical Analysis

Adult wings, mean fluorescence intensity, and cell counts were measured using Fiji. GraphPad Prism 10.0 was used for statistical analysis and graphical representation. Graphs depict the mean of each treatment, while error bars represent the standard deviation. The mean fluorescence intensity of EdU labeling was quantified in Fiji. In each experiment, the mean fluorescence intensity (MFI) of the wing pouch was normalized to the MFI of the entire disc. The mean and standard deviation for each normalized treatment was calculated and used for statistical analysis. The sample size and P values for all statistical analyses are indicated in the figure legends. Statistical significance was evaluated in Prism 10.0 using a Student's T-test or a one-way ANOVA with a multiple comparisons test.

Data and reagent availability

Stocks are available upon request and details of stocks and reagents used in this study are available in the materials and methods. The authors affirm that all data necessary for confirming the conclusions of the article are present within the article, figures, and tables.

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Author contributions

Jacob Klemm, Conceptualization, Investigation Data curation, Formal analysis, Validation, Writing – original draft, Writing – review and editing; Chloe Van Hazel, Resources; Robin Harris, Supervision, Funding acquisition, Investigation, Resources, Writing – review and editing.

Additional files

Supplementary Genotypes 

References

1. Amcheslavsky A., Wang S., Fogarty C. E., Lindblad J. L., Fan Y., Bergmann A. 2018) **Plasma Membrane Localization of Apoptotic Caspases for Non-apoptotic Functions** *Dev Cell* **45**:450–464
2. Anthoney N., Foldi I., Hidalgo A. 2018) **Toll and Toll-like receptor signalling in development** *Development* **145**:9
3. Ashkenazi A., Salvesen G. 2014) **Regulated cell death: signaling and mechanisms** *Annu Rev Cell Dev Biol* **30**:337–356
4. Bach E. A., Ekas L. A., Ayala-Camargo A., Flaherty M. S., Lee H., Perrimon N., Baeg G. H. 2007) **GFP reporters detect the activation of the Drosophila JAK/STAT pathway in vivo** *Gene Expr Patterns* **7**:323–331
5. Baena-Lopez L. A., Arthurton L., Bischoff M., Vincent J. P., Alexandre C., McGregor R. 2018) **Novel initiator caspase reporters uncover previously unknown features of caspase-activating cells** *Development* **145**:23
6. Bardet P. L., Kolahgar G., Mynett A., Miguel-Aliaga I., Briscoe J., Meier P., Vincent J. P. 2008) **A fluorescent reporter of caspase activity for live imaging** *Proc Natl Acad Sci U S A* **105**:13901–13905
7. Beira J. V., Paro R. 2016) **The legacy of Drosophila imaginal discs** *Chromosoma* **125**:573–592
8. Bergantinos C., Corominas M., Serras F. 2010) **Cell death-induced regeneration in wing imaginal discs requires JNK signalling** *Development* **137**:1169–1179
9. Bergmann A., Steller H. 2010) **Apoptosis, stem cells, and tissue regeneration** *Sci Signal* **3**:re8
10. Bonne S. L., Kadri S. S. 2017) **Evaluation and Management of Necrotizing Soft Tissue Infections** *Infect Dis Clin North Am* **31**:497–511
11. Brown-Suedel A. N., Bouchier-Hayes L. 2020) **Caspase-2 Substrates: To Apoptosis, Cell Cycle Control, and Beyond** *Front Cell Dev Biol* **8**:610022
12. Cao X 2016) **Self-regulation and cross-regulation of pattern-recognition receptor signalling in health and disease** *Nat Rev Immunol* **16**:35–50
13. Chatterjee N., Bohmann D. 2012) **A versatile PhiC31 based reporter system for measuring AP-1 and Nrf2 signaling in Drosophila and in tissue culture** *PLoS One* **7**:e34063
14. Chera S., Ghila L., Dobretz K., Wenger Y., Bauer C., Buzgariu W., Martinou J. C., Galliot B. 2009) **Apoptotic cells provide an unexpected source of Wnt3 signaling to drive hydra head regeneration** *Dev Cell* **17**:279–289
15. Colon Plaza S., Su T. T. 2024) **Ionizing radiation induces cells with past caspase activity that contribute to the adult organ in Drosophila and show reduced Loss of Heterozygosity** *Cell Death Discov* **10**:6

16. D'Arcy M. S 2019) **Cell death: a review of the major forms of apoptosis, necrosis and autophagy** *Cell Biol Int* **43**:582–592
17. Ding A. X., Sun G., Argaw Y. G., Wong J. O., Easwaran S., Montell D. J. 2016) **CasExpress reveals widespread and diverse patterns of cell survival of caspase-3 activation during development in vivo** *eLife* **5**
18. Diwanji N., Bergmann A. 2018) **An unexpected friend - ROS in apoptosis-induced compensatory proliferation: Implications for regeneration and cancer** *Semin Cell Dev Biol* **80**:74–82
19. Evans C. J., Olson J. M., Ngo K. T., Kim E., Lee N. E., Kuoy E., Patananan A. N., Sitz D., Tran P., Do M. T., Yackle K., Cespedes A., Hartenstein V., Call G. B., Banerjee U. 2009) **G-TRACE: rapid Gal4-based cell lineage analysis in Drosophila** *Nat Methods* **6**:603–605
20. Fan Y., Bergmann A. 2008) **Apoptosis-induced compensatory proliferation. The Cell is dead. Long live the Cell!** *Trends Cell Biol* **18**:467–473
21. Fan Y., Bergmann A. 2008) **Distinct mechanisms of apoptosis-induced compensatory proliferation in proliferating and differentiating tissues in the Drosophila eye** *Dev Cell* **14**:399–410
22. Fan Y., Wang S., Hernandez J., Yenigun V. B., Hertlein G., Fogarty C. E., Lindblad J. L., Bergmann A. 2014) **Genetic models of apoptosis-induced proliferation decipher activation of JNK and identify a requirement of EGFR signaling for tissue regenerative responses in Drosophila** *PLoS Genet* **10**:e1004131
23. Festjens N., Vanden Berghe T., Vandenabeele P. 2006) **Necrosis, a well-orchestrated form of cell demise: signalling cascades, important mediators and concomitant immune response** *Biochim Biophys Acta* **1757**:1371–1387
24. Florentin A., Arama E. 2012) **Caspase levels and execution efficiencies determine the apoptotic potential of the cell** *J Cell Biol* **196**:513–527
25. Fogarty C. E., Bergmann A. 2017) **Killers creating new life: caspases drive apoptosis-induced proliferation in tissue repair and disease** *Cell Death Differ* **24**:1390–1400
26. Fogarty C. E., Diwanji N., Lindblad J. L., Tare M., Amcheslavsky A., Makhijani K., Bruckner K., Fan Y., Bergmann A. 2016) **Extracellular Reactive Oxygen Species Drive Apoptosis-Induced Proliferation via Drosophila Macrophages** *Curr Biol* **26**:575–584
27. Fox D. T., Cohen E., Smith-Bolton R. 2020) **Model systems for regeneration: Drosophila** *Development* **147**:7
28. Fuchs Y., Steller H. 2015) **Live to die another way: modes of programmed cell death and the signals emanating from dying cells** *Nat Rev Mol Cell Biol* **16**:329–344
29. Gong T., Liu L., Jiang W., Zhou R. 2020) **DAMP-sensing receptors in sterile inflammation and inflammatory diseases** *Nat Rev Immunol* **20**:95–112
30. Gordon O., Henry C. M., Srinivasan N., Ahrens S., Franz A., Deddouche S., Chakravarty P., Phillips D., George R., Kjaer S., Frith D., Snijders A. P., Valente R. S., Simoes da Silva C. J., Teixeira L., Thompson B., Dionne M. S., Wood W., Reis e Sousa C. 2018) **alpha-actinin accounts for the**

bioactivity of actin preparations in inducing STAT target genes in *Drosophila melanogaster* *eLife* 7

31. Grasl-Kraupp B., Ruttkay-Nedecky B., Koudelka H., Bukowska K., Bursch W., Schulte-Hermann R. 1995) **In situ detection of fragmented DNA (TUNEL assay) fails to discriminate among apoptosis, necrosis, and autolytic cell death: a cautionary note** *Hepatology* **21**:1465–1468
32. Hajibabaei F., Abedpoor N., Mohamadynejad P. 2023) **Types of Cell Death from a Molecular Perspective** *Biology (Basel)* **12**:11
33. Hakkarainen T. W., Kopari N. M., Pham T. N., Evans H. L. 2014) **Necrotizing soft tissue infections: review and current concepts in treatment, systems of care, and outcomes** *Curr Probl Surg* **51**:344–362
34. Hariharan I. K., Serras F. 2017) **Imaginal disc regeneration takes flight** *Curr Opin Cell Biol* **48**:10–16
35. Harris R. E 2023) **Investigating Tissue Regeneration Using the DUAL Control Genetic Ablation System** *Methods Mol Biol* **2599**:255–270
36. Harris R. E., Stinchfield M. J., Nystrom S. L., McKay D. J., Hariharan I. K. 2020) **Damage-responsive, maturity-silenced enhancers regulate multiple genes that direct regeneration in *Drosophila*** *eLife* **9**:e58305
37. Hawkins C. J., Yoo S. J., Peterson E. P., Wang S. L., Vernooy S. Y., Hay B. A. 2000) **The *Drosophila* caspase DRONC cleaves following glutamate or aspartate and is regulated by DIAP1, HID, and GRIM** *J Biol Chem* **275**:27084–27093
38. Hay B. A., Wolff T., Rubin G. M. 1994) **Expression of baculovirus P35 prevents cell death in *Drosophila*** *Development* **120**:2121–2129
39. Herrera S. C., Bach E. A. 2019) **JAK/STAT signaling in stem cells and regeneration: from *Drosophila* to vertebrates** *Development* **146**:2
40. Huh J. R., Guo M., Hay B. A. 2004) **Compensatory proliferation induced by cell death in the *Drosophila* wing disc requires activity of the apical cell death caspase Dronc in a nonapoptotic role** *Curr Biol* **14**:1262–1266
41. Jaiswal J., Egert J., Engesser R., Peyroton A. A., Nogay L., Weichselberger V., Crucianelli C., Grass I., Kreutz C., Timmer J., Classen A. K. 2023) **Mutual repression between JNK/AP-1 and JAK/STAT stratifies senescent and proliferative cell behaviors during tissue regeneration** *PLoS Biol* **21**:e3001665
42. Jung Y., Witek R. P., Syn W. K., Choi S. S., Omenetti A., Premont R., Guy C. D., Diehl A. M. 2010) **Signals from dying hepatocytes trigger growth of liver progenitors** *Gut* **59**:655–665
43. Kamber Kaya H. E., Ditzel M., Meier P., Bergmann A. 2017) **An inhibitory mono-ubiquitylation of the *Drosophila* initiator caspase Dronc functions in both apoptotic and non-apoptotic pathways** *PLoS Genet* **13**:e1006438
44. Karsch-Bluman A., Feiglin A., Arbib E., Stern T., Shoval H., Schwob O., Berger M., Benny O. 2019) **Tissue necrosis and its role in cancer progression** *Oncogene* **38**:1920–1935

45. Katsuyama T., Comoglio F., Seimiya M., Cabuy E., Paro R. 2015) **During *Drosophila* disc regeneration, JAK/STAT coordinates cell proliferation with Dilp8-mediated developmental delay** *Proc Natl Acad Sci U S A* **112**:E2327–2336
46. Khan S. J., Abidi S. N. F., Skinner A., Tian Y., Smith-Bolton R. K. 2017) **The *Drosophila* Duox maturation factor is a key component of a positive feedback loop that sustains regeneration signaling** *PLoS Genet* **13**:e1006937
47. Kim J., Sebring A., Esch J. J., Kraus M. E., Vorwerk K., Magee J., Carroll S. B. 1996) **Integration of positional signals and regulation of wing formation and identity by *Drosophila* vestigial gene** *Nature* **382**:133–138
48. Klemm J. M. J., Stinchfield, Harris R. E. 2021) **Necrosis-induced apoptosis promotes regeneration in *Drosophila* wing imaginal discs** *Genetics* **219**
49. Konstantinidis K., Whelan R. S., Kitsis R. N. 2012) **Mechanisms of cell death in heart disease** *Arterioscler Thromb Vasc Biol* **32**:1552–1562
50. La Fortezza M., Schenk M., Cosolo A., Kolybaba A., Grass I., Classen A. K. 2016) **JAK/STAT signalling mediates cell survival in response to tissue stress** *Development* **143**:2907–2919
51. La Marca J. E., Richardson H. E. 2020) **Two-Faced: Roles of JNK Signalling During Tumourigenesis in the *Drosophila* Model** *Front Cell Dev Biol* **8**:42
52. Ledru M., Clark C. A., Brown J., Verghese S., Ferrara S., Goodspeed A., Su T. T. 2022) **Differential gene expression analysis identified determinants of cell fate plasticity during radiation-induced regeneration in *Drosophila*** *PLoS Genet* **18**:e1009989
53. Li F., Huang Q., Chen J., Peng Y., Roop D. R., Bedford J. S., Li C. Y. 2010) **Apoptotic cells activate the “phoenix rising” pathway to promote wound healing and tissue regeneration** *Sci Signal* **3**:ra13
54. Li M., Sun S., Priest J., Bi X., Fan Y. 2019) **Characterization of TNF-induced cell death in *Drosophila* reveals caspase- and JNK-dependent necrosis and its role in tumor suppression** *Cell Death Dis* **10**:613
55. Li Z., Shao W., Lv X., Wang B., Han L., Gong S., Wang P., Feng Y. 2023) **Advances in experimental models of osteonecrosis of the femoral head** *J Orthop Translat* **39**:88–99
56. Liu K., Li Y., Liu L. 2013) **Modeling calcium-overload mediated necrosis in *Drosophila*** *Methods Mol Biol* **1004**:203–213
57. Madigan J. P., Chotkowski H. L., Glaser R. L. 2002) **DNA double-strand break-induced phosphorylation of *Drosophila* histone variant H2Av helps prevent radiation-induced apoptosis** *Nucleic Acids Res* **30**:3698–3705
58. Martin R., Morata G. 2018) **Regenerative response of different regions of *Drosophila* imaginal discs** *Int J Dev Biol* **62**:507–512
59. Martin R., Pinal N., Morata G. 2017) **Distinct regenerative potential of trunk and appendages of *Drosophila* mediated by JNK signalling** *Development* **144**:3946–3956
60. Masi L., Falchetti A., Brandi M. L. 2007) **Osteonecrosis in genetic disorders** *Clin Cases Miner Bone Metab* **4**:27–29

61. Meier P., Silke J., Leever S. J., Evan G. I. 2000) **The Drosophila caspase DRONC is regulated by DIAP1** *EMBO J* **19**:598–611
62. Ming M., Obata F., Kuranaga E., Miura M. 2014) **Persephone/Spatzle pathogen sensors mediate the activation of Toll receptor signaling in response to endogenous danger signals in apoptosis-deficient Drosophila** *J Biol Chem* **289**:7558–7568
63. Monier B., Gettings M., Gay G., Mangeat T., Schott S., Guarner A., Suzanne M. 2015) **Apico-basal forces exerted by apoptotic cells drive epithelium folding** *Nature* **518**:245–248
64. Morata G., Shlevkov E., Perez-Garijo A. 2011) **Mitogenic signaling from apoptotic cells in Drosophila** *Dev Growth Differ* **53**:168–176
65. Mulay S. R., Linkermann A., Anders H. J. 2016) **Necroinflammation in Kidney Disease** *J Am Soc Nephrol* **27**:27–39
66. Nakada H., Inoue M., Furuya K., Watanabe H., Ikegame K., Nakayama Y., Ohmori M., Nakagomi H. 2019) **Fat necrosis after breast-conserving oncoplastic surgery** *Breast Cancer* **26**:125–130
67. Nano M., Montell D. J. 2024) **Apoptotic signaling: Beyond cell death** *Semin Cell Dev Biol* **156**:22–34
68. Nishida H., Alberio A. B., Onoue K., Ikegawa Y., Sulekh S., Sakizli U., Minami Y., Yonemura S., Wang Y. C., Yoo S. K. 2024) **Necrosensor: a genetically encoded fluorescent sensor for visualizing necrosis in Drosophila** *Biol Open* **13**
69. Pellettieri J., Fitzgerald P., Watanabe S., Mancuso J., Green D. R., Sanchez Alvarado A. 2010) **Cell death and tissue remodeling in planarian regeneration** *Dev Biol* **338**:76–85
70. Perez-Garijo A 2018) **When dying is not the end: Apoptotic caspases as drivers of proliferation** *Semin Cell Dev Biol* **82**:86–95
71. Perez-Garijo A., Martin F. A., Morata G. 2004) **Caspase inhibition during apoptosis causes abnormal signalling and developmental aberrations in Drosophila** *Development* **131**:5591–5598
72. Perez-Garijo A., Martin F. A., Struhl G., Morata G. 2005) **Dpp signaling and the induction of neoplastic tumors by caspase-inhibited apoptotic cells in Drosophila** *Proc Natl Acad Sci U S A* **102**:17664–17669
73. Perez-Garijo A., Shlevkov E., Morata G. 2009) **The role of Dpp and Wg in compensatory proliferation and in the formation of hyperplastic overgrowths caused by apoptotic cells in the Drosophila wing disc** *Development* **136**:1169–1177
74. Perez-Garijo A., Steller H. 2015) **Spreading the word: non-autonomous effects of apoptosis during development, regeneration and disease** *Development* **142**:3253–3262
75. Pinal N., Calleja M., Morata G. 2019) **Pro-apoptotic and pro-proliferation functions of the JNK pathway of Drosophila: roles in cell competition, tumorigenesis and regeneration** *Open Biol* **9**:180256
76. Pinal N., Martín M., Medina I., Morata G. 2018) **Short-term activation of the Jun N-terminal kinase pathway in apoptosis-deficient cells of Drosophila induces tumorigenesis** *Nature Communications* **9**:1

77. Robertson S. A., Jeevaratnam J. A., Agrawal A., Cutress R. I. 2017) **Mastectomy skin flap necrosis: challenges and solutions** *Breast Cancer (Dove Med Press)* **9**:141–152
78. Roh J. S., Sohn D. H. 2018) **Damage-Associated Molecular Patterns in Inflammatory Diseases** *Immune Netw* **18**:e27
79. Ryoo H. D., Bergmann A. 2012) **The role of apoptosis-induced proliferation for regeneration and cancer** *Cold Spring Harb Perspect Biol* **4**:a008797
80. Ryoo H. D., Gorenc T., Steller H. 2004) **Apoptotic cells can induce compensatory cell proliferation through the JNK and the Wingless signaling pathways** *Dev Cell* **7**:491–501
81. Schott S., Ambrosini A., Barbaste A., Benassayag C., Gracia M., Proag A., Rayer M., Monier B., Suzanne M. 2017) **A fluorescent toolkit for spatiotemporal tracking of apoptotic cells in living Drosophila tissues** *Development* **144**:3840–3846
82. Shapiro P. J., Hsu H. H., Jung H., Robbins E. S., Ryoo H. D. 2008) **Regulation of the Drosophila apoptosome through feedback inhibition** *Nat Cell Biol* **10**:1440–1446
83. Shinoda N., Hanawa N., Chihara T., Koto A., Miura M. 2019) **Dronc-independent basal executioner caspase activity sustains Drosophila imaginal tissue growth** *Proc Natl Acad Sci U S A* **116**:20539–20544
84. Siegrist S. E., Haque N. S., Chen C. H., Hay B. A., Hariharan I. K. 2010) **Inactivation of both Foxo and reaper promotes long-term adult neurogenesis in Drosophila** *Curr Biol* **20**:643–648
85. Su T. T 2020) **Non-apoptotic roles of apoptotic proteases: new tricks for an old dog** *Open Biol* **10**:200130
86. Tonnus W., Belavgeni A., Beuschlein F., Eisenhofer G., Fassnacht M., Kroiss M., Krone N. P., Reincke M., Bornstein S. R., Linkermann A. 2021) **The role of regulated necrosis in endocrine diseases** *Nat Rev Endocrinol* **17**:497–510
87. Tseng A. S., Adams D. S., Qiu D., Koustubhan P., Levin M. 2007) **Apoptosis is required during early stages of tail regeneration in Xenopus laevis** *Dev Biol* **301**:62–69
88. Venereau E., Ceriotti C., Bianchi M. E. 2015) **DAMPs from Cell Death to New Life** *Front Immunol* **6**:422
89. Verghese S., Su T. T. 2016) **Drosophila Wnt and STAT Define Apoptosis-Resistant Epithelial Cells for Tissue Regeneration after Irradiation** *PLoS Biol* **14**:e1002536
90. Verghese S., Su T. T. 2018) **Ionizing radiation induces stem cell-like properties in a caspase-dependent manner in Drosophila** *PLoS Genet* **14**:e1007659
91. Vríz S., Reiter S., Galliot B. 2014) **Cell death: a program to regenerate** *Curr Top Dev Biol* **108**:121–151
92. Wang L. H., Baker N. E. 2019) **Salvador-Warts-Hippo pathway regulates sensory organ development via caspase-dependent nonapoptotic signaling** *Cell Death Dis* **10**:669
93. Worley M. I., Hariharan I. K. 2022) **Imaginal Disc Regeneration: Something Old, Something New** *Cold Spring Harb Perspect Biol* **14**:11

- 94. Worley M. I., Setiawan L., Hariharan I. K. 2012) **Regeneration and trans determination in *Drosophila* imaginal discs** *Annu Rev Genet* **46**:289–310
- 95. Wu M. Y., Yiang G. T., Liao W. T., Tsai A. P., Cheng Y. L., Cheng P. W., Li C. Y., Li C. J. 2018) **Current Mechanistic Concepts in Ischemia and Reperfusion Injury** *Cell Physiol Biochem* **46**:1650–1667
- 96. Xu D., Li Y., Arcaro M., Lackey M., Bergmann A. 2005) **The CARD-carrying caspase Dronc is essential for most, but not all, developmental cell death in *Drosophila*** *Development* **132**:2125–2134
- 97. Yang X., Ren H., Fu J. 2021) **Treatment of Radiation-Induced Brain Necrosis** *Oxid Med Cell Longev* **2021**:4793517
- 98. Yang Y., Hou L., Li Y., Ni J., Liu L. 2013) **Neuronal necrosis and spreading death in a *Drosophila* genetic model** *Cell Death Dis* **4**:e723

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Reviewer #2 (Public review):

In this revised manuscript, Klemm et al., build on top of past published findings (Klemm et al., 2021) to characterize caspase activation in distal cells following necrotic tissue damage within the *Drosophila* wing imaginal disc. Previously in Klemm et al., 2021, the authors describe necrosis-induced-apoptosis (NiA) following the development of a genetic system to study necrosis that is caused by the expression of a constitutive active GluR1 (Glutamate/Ca²⁺

channel), and they discovered that the appearance of NiA cells were important for promoting regeneration.

In this manuscript, the authors investigate how tissues regenerate following necrotic cell death. They find that:

- (1) the cells of the wing pouch are more likely to have non-autonomous caspase activation than other regions within the wing imaginal disc (hinge and notum),
- (2) two signaling pathways that are known to be upregulated during regeneration, Wnt (wingless) and JAK/Stat signaling, act to prevent additional NiA in pouch cells, and may partially explain the region specificity,
- (3) the presence of NiA (and/or NiCP) cells promotes regenerative proliferation in the late stages of regeneration,
- (4) not all caspase-positive cells are cleared from the epithelium (these cells are then referred to as Necrosis-induced Caspase Positive (NiCP) cells), these NiCP cells continue to live and promote proliferation in adjacent cells,
- (5) the initiator caspase Dronc is important for creating NiA/NiCP cells and for these cells to promote proliferation. Animals heterozygous for a Dronc null allele show a decrease in regeneration following necrotic tissue damage. In the revised manuscript, the authors provide improvements through additional data quantifications and text changes to better explain NiA/NiCP lineage tracing methods.

The study has the potential to be broadly interesting due to the insights into how tissues differentially respond to necrosis as compared to apoptosis to promote regeneration. The paper raises many interesting questions for future investigation, including what is the nature of the signaling between the damaged tissue and the NiA/NiCP responsive areas (such as the identity of the DAMPs)? What determines if these cells at a distance undergo apoptosis or remain viable in the tissue as caspase-positive cells? And since the authors have data that indicates that the phenomenon is distinct from 'undead cells', what are the mechanisms by which these cells promote local proliferation?

<https://doi.org/10.7554/eLife.101114.2.sa2>

Reviewer #3 (Public review):

The manuscript "Regeneration following tissue necrosis is mediated by non-apoptotic caspase activity" by Klemm et al. is an exploration of what happens to a group of cells that experience caspase activation after necrosis occurs some distance away from the cells of interest. These experiments have been conducted in the *Drosophila* wing imaginal disc, which has been used extensively to study the response of a developing epithelium to damage and stress. The authors revise and refine their earlier discovery of apoptosis initiated by necrosis, here showing that many of those presumed apoptotic cells do not complete apoptosis. Thus, the most interesting aspect of the paper is the characterization of a group of cells that experience mild caspase activation in response to an unknown signal, followed by some effector caspase activation and DNA damage, but that then recover from the DNA damage, avoid apoptosis, and proliferate instead.

The authors have addressed the concerns raised, including those about drawing conclusions from RNAi knockdown without evaluating the efficacy of the knockdown, and in doing so they revised their conclusions after ascertaining that the Zfh2 RNAi was not effective.

The authors have added quantification of the imaging data throughout, which strengthens their conclusions.

In addition, the authors have revised some of the text describing the changes in EdU signal and added explanations of reagents such as the caspase sensors to clarify the experimental approaches, results, and interpretation of those results.

The authors have also addressed the minor concerns and questions about the figures and text.

A few questions remain, which the authors may choose to address.

(1) The hh>Stat92ERNAi was assessed by the 10xSTAT-GFP reporter, as shown in Fig 2 Supp1 F. The authors point out the marked reduction in GFP in the ventral part of the hinge but do not comment on the lack of change in GFP in the dorsal part of the hinge. However, the open arrowhead in Figure 2H indicating the lack of cDcp-1 signal in the hinge in the same experiment points to the dorsal hinge, where the reporter suggests no difference in JAK-STAT signaling.

(2) The data used to conclude that DRONC-DN and UAS-DIAP1 do not affect regenerative proliferation were normalized EdU intensities. As discussed in the prior review round, normalized EdU may not be a good comparison across experimental conditions given that the remainder of the disc may also have altered EdU incorporation, so this measurement may not be enough by itself to draw conclusions about regenerative proliferation. To strengthen the conclusion that regenerative proliferation is unaffected under these conditions, the authors may want to consider using a second measure such as adult wing size, PCNA, or quantitate mitoses via anti-phospho histone H3 staining.

<https://doi.org/10.7554/eLife.101114.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

In previous work, the authors described necrosis-induced apoptosis (NiA) as a consequence of induced necrosis. Specifically, experimentally induced necrosis in the distal pouch of larval wing imaginal discs triggers NiA in the lateral pouch. In this manuscript, the authors confirmed this observation and found that while necrosis can kill all areas of the disc, NiA is limited to the pouch and to some extent to the notum, but is excluded from the hinge region. Interestingly and unexpectedly, signaling by the Jak/Stat and Wg pathways inhibits NiA. Further characterization of NiA by the authors reveals that NiA also triggers regenerative proliferation which can last up to 64 hours following necrosis induction. This regenerative response to necrosis is significantly stronger compared to discs ablated by apoptosis. Furthermore, the regenerative proliferation induced by necrosis is dependent on the apoptotic pathway because RNAi targeting the RHG genes is sufficient to block proliferation. However, NiA does not promote proliferation through the previously described apoptosis-induced proliferation (AiP) pathway, although cells at the wound edge undergo AiP. Further examination of the caspase levels in NiA cells allowed the authors to group these cells into two clusters: some cells (NiA) undergo apoptosis and are removed, while others referred to as Necrosis-induced Caspase Positive (NiCP) cells survive despite caspase activity. It is the

*NiCP cells that repair cellular damage including DNA damage and that promote regenerative proliferation. Caspase sensors demonstrate that both groups of cells have initiator caspase activity, while only the NiA cells contain effector caspase activity. Under certain conditions, the authors were also able to visualize effector caspase activity in NiCP cells, but the level was low, likely below the threshold for apoptosis. Finally, the authors found that loss of the initiator caspase *Dronc* blocks regenerative proliferation, while inhibiting effector caspases by expression of *p35* does not, suggesting that *Dronc* can induce regenerative proliferation following necrosis in a non- apoptotic manner. This last finding is very interesting as it implies that *Dronc* can induce proliferation in at least two ways in addition to its requirement in AiP.*

Strengths:

*This is a very interesting manuscript. The authors demonstrate that epithelial tissue that contains a significant number of necrotic cells is able to regenerate. This regenerative response is dependent on the apoptotic pathway which is induced at a distance from the necrotic cells. Although regenerative proliferation following necrosis requires the initiator caspase *Dronc*, *Dronc* does not induce a classical AiP response for this type of regenerative response. In future work, it will be very interesting to dissect this regenerative response pathway genetically.*

Weaknesses:

No weaknesses were identified.

We thank the reviewer for their positive evaluation and kind words.

Reviewer #2 (Public Review):

Summary / Strengths:

*In this manuscript, Klemm et al., build on past published findings (Klemm et al., 2021) to characterize caspase activation in distal cells following necrotic tissue damage within the *Drosophila* wing imaginal disc. Previously in Klemm et al., 2021, the authors describe necrosis-induced-apoptosis (NiA) following the development of a genetic system to study necrosis that is caused by the expression of a constitutive active *GluR1* (Glutamate/Ca²⁺ channel), and they discovered that the appearance of NiA cells were important for promoting regeneration.*

*In this manuscript, the authors aim to investigate how tissues regenerate following necrotic cell death. They find that the cells of the wing pouch are more likely to have non-autonomous caspase activation than other regions within the wing imaginal disc (hinge and notum), two signaling pathways that are known to be upregulated during regeneration, Wnt (wingless) and JAK/Stat signaling, act to prevent additional NiA in pouch cells, and may explain the region specificity, the presence of NiA cells promotes regenerative proliferation in late stages of regeneration, not all caspase-positive cells are cleared from the epithelium (these cells are then referred to as Necrosis-induced Caspase Positive (NiCP) cells), these NiCP cells continue to live and promote proliferation in adjacent cells, the caspase *Dronc* is important for creating NiA/NiCP cells and for these cells to promote proliferation. Animals heterozygous for a *Dronc* null allele show a decrease in regeneration following necrotic tissue damage.*

The study has the potential to be broadly interesting due to the insights into how tissues differentially respond to necrosis as compared to apoptosis to promote regeneration.

Weaknesses:

However, here are some of my current concerns for the manuscript in its current version:

The presence of cells with activated caspase that don't die (NiCP cells) is an interesting biological phenomenon but is not described until Figure 5. How does the existence of NiCP cells impact the earlier findings presented? Is late proliferation due to NiA, NiCP, or both? Does Wg and JAK/STAT signaling act to prevent the formation of both NiA and NiCP cells or only NiA cells? Moreover, the authors are able to specifically manipulate the wound edge (WE) and lateral pouch cells (LP), but don't show how these manipulations within these distinct populations impact regeneration. The authors provide evidence that driving UAS-mir(RHG) throughout the pouch, in the LP or the WE all decrease the amount of NiA/NiCP in Figure 3G-O, but no data on final regenerative outcomes for these manipulations is presented (such as those presented for Dronc-/- in Fig 7M). The manuscript would be greatly enhanced by quantification of more of the findings, especially in describing if the specific manipulations that impacted NiA /NiCP cells disrupt end-point regeneration phenotypes.

We have added a line to the results to clarify that we believe the finding that some NiA likely persist as NiCP does not affect our conclusions up to this point.

We have added a statement emphasizing the results from our first paper, which demonstrate that LP>miRHG expression reduces the overall capacity to regenerate.

Quantification of the change in posterior NiA number have been added to Figure 2L to strengthen the evidence. Likewise, we have included quantification of the E2F time course presented in Figure 3A (Figure 3 – Figure supplement 1C), and quantification of the change in GC3Ai signal over time has been added to Figure 5 - Figure supplement 1D) to emphasize the perdurance of GC3Ai-positive NiA/NiCP.

How fast does apoptosis take within the wing disc epithelium? How many of the caspase(+) cells are present for the whole 48 hours of regeneration? Are new cells also induced to activate caspase during this time window? The author presented a number of interesting experiments characterizing the NiCP cells. For the caspase sensor GC3Ai experiments in Figure 5, is there a way to differentiate between cells that have maintained fluorescent GC3Ai from cells that have newly activated caspase? What is the timeline for when NiA and NiCP are specified? In addition, what fraction of NiCP cells contribute to the regenerated epithelium? Additional information about the temporal dynamics of NiA and NiCP specification/commitment would be greatly appreciated.

We have included more information concerning the kinetics of apoptotic cell removal, and how this compares to the observations we have made with NiA/NiCP in our GC3Ai experiments. Additionally, we have included a quantification of the percent of the whole wing pouch with GC3Ai signal over time (Figure 5F) as well as the distal wing pouch with GC3Ai signal over time (Figure 5 – Figure supplement 1D) to further support the idea that NiCP persist over time.

We acknowledge that our GC3Ai time course unfortunately cannot confirm whether the increase in GC3Ai signal over time is due to cells with new caspase activity or proliferating NiCP and have included this point in the discussion.

We attempted to track the lineage of NiA/NiCP into the pupal and adult wings with CasExpress and DBS, however the results of these experiments were inconsistent, and therefore we did not feel confident to include these data or draw conclusions in either direction. We are currently designing variations of these lineage trace tools in order to better track the lineage of these cells that we hope to include in a future paper.

The notum also does not express developmental JAK/STAT, yet little NiA was observed within the notum. Do the authors have any additional insights into the differential response between the pouch and notum? What makes the pouch unique? Are NiA/NiCP cells created within other imaginal discs and other tissues? Are they similarly important for regenerative responses in other contexts?

We have added a brief mention of these points to the appropriate results section to avoid further increasing the length of the discussion.

Data on the necrosis of other imaginal discs through FLP/FRT clone formation in haltere and leg discs has been added to Figure 1 Figure supplement 1J, and described in the text.

Reviewer #3 (Public Review):

The manuscript "Regeneration following tissue necrosis is mediated by non- apoptotic caspase activity" by Klemm et al. is an exploration of what happens to a group of cells that experience caspase activation after necrosis occurs some distance away from the cells of interest. These experiments have been conducted in the Drosophila wing imaginal disc, which has been used extensively to study the response of a developing epithelium to damage and stress. The authors revise and refine their earlier discovery of apoptosis initiated by necrosis, here showing that many of those presumed apoptotic cells do not complete apoptosis. Thus, the most interesting aspect of the paper is the characterization of a group of cells that experience mild caspase activation in response to an unknown signal, followed by some effector caspase activation and DNA damage, but that then recover from the DNA damage, avoid apoptosis, and proliferate instead. Many questions remain unanswered, including the signal that stimulates the mild caspase activation, and the mechanism through which this activation stimulates enhanced proliferation.

The authors should consider answering additional questions, clarifying some points, and making some minor corrections:

Major concerns affecting the interpretation of experimental results:

Expression of STAT92E RNAi had no apparent effect on the ability of hinge cells to undergo NiA, leading the authors to conclude that other protective signals must exist. However, the authors have not shown that this STAT92E RNAi is capable of eliminating JAK/STAT signaling in the hinge under these experimental conditions. Using a reporter for JAK/STAT signaling, such as the STAT-GFP, as a readout would confirm the reduction or elimination of signaling. This confirmation would be necessary to support the negative result as presented.

We have included data demonstrating our ability to knock down JAK/STAT activity in the hinge with UAS-Stat92E^{RNAi} (Figure 2 – Figure supplement 1E and F). Additionally, we have included a quantification of posterior NiA/NiCP with the Stat92E^{RNAi} (as well as wg^{RNAi} and Zfh-2^{RNAi}, Figure 2L) to strengthen our conclusion that JAK/STAT and WNT signaling acts to regulate NiA formation within the pouch.

Similarly, the authors should confirm that the Zfh2 RNAi is reducing or eliminating Zfh2 levels in the hinge under these experimental conditions, before concluding that Zfh2 does not play a role in stopping hinge cells from undergoing NiA.

We have repeated this experiment with a longer knockdown using a GAL4 driver that expresses from early larval stages until our evaluation at L3, but were unable to demonstrate a loss of Zfh-2 with IF labeling. Additionally, we have quantified posterior NiA/NiCP with a

Zfh-2RNAi (Figure 2L) and do find a slight increase in NiA/NiCP number, however this change is not significant. We have altered our conclusions to reflect these new data.

EdU incorporation was quantified by measuring the fluorescence intensity of the pouch and normalizing it to the fluorescence intensity of the whole disc. However, the images show that EdU fluorescence intensity of other regions of the disc, especially the notum, varied substantially when comparing the different genetic backgrounds (for example, note the substantially reduced EdU in the notum of Figure 3 B' and B'). Indeed, it has been shown that tissue damage can lead to suppression of proliferation in the notum and elsewhere in the disc, unless the signaling that induces the suppression is altered. Therefore, the normalization may be skewing the results because the notum EdU is not consistent across samples, possibly because the damage-induced suppression of proliferation in the notum is different across the different genetic backgrounds.

To more accurately reflect the observations that we have made with the EdU assay, we have changed our terminology to indicate that the EdU signal is more localized to the damaged tissue in ablated discs, thus taking into account the relative changes across the disc, rather than referring to it as an increase in the pouch. To further strengthen our observation that damage results in a localized proliferation, we have included a quantification of the E2F time course presented in Figure 3A (Figure 3 – Figure supplement 1C), which underscores the trend observed in our EdU experiments.

The authors expressed p35 to attempt to generate "undead cells". They take an absence of mitogen secretion or increased proliferation as evidence that undead cells were not generated. However, there could be undead cells that do not stimulate proliferation non-autonomously, which could be detected by the persistence of caspase activity in cells that do not complete apoptosis. Indeed, expressing p35 and observing sustained effector caspase activation could help answer the later question of what percentage of this cell population would otherwise complete apoptosis (NiA, rescued by p35) vs reverse course and proliferate (NiCP, unaffected by p35).

In our previous work, we showed that P35 expression impairs our ability to detect effector caspases with IF-based tools. This can also be seen in Figure 4 of this work (Figure 4C and F). Given that P35 expression precludes our ability to label and assay effector caspase activity visually, and thus address the concerns outlined above, we relied on other tools such as reporters of AiP mitogens (*wg-lacZ* & *dpp-lacZ*) to assay whether NiA participate in AiP. As a functional readout, we also paired P35 expression with the EdU assay to test whether proliferation was altered by the presence of undead cells. The results discussed in Figure 4 lead us to conclude that NiA likely do not participate in the canonical AiP feedforward loop, although it is possible that these experiments generate another type of undead cell – one that utilizes a different mechanism to promote proliferation.

It is unclear if the authors' model is that the NiCP cells lead to autonomous or non-autonomous cell proliferation, or both. Could the lineage-tracing experiments and/or the experiments marking mitosis relative to caspase activity answer this question?

We have added further details to the discussion on the potential for NiA/NiCP to induce cell autonomous/non-autonomous proliferation.

Many of the conclusions rely on single images. Quantification of many samples should be included wherever possible.

We have added quantification to strengthen the results of Figures 2, 3 and 5.

Why does the reduction of Dronc appear to affect regenerative growth in females but not males?

We have repeated this regeneration scoring experiments and have increased the N for control versus *droncI29* mutant males, however the results of the analysis for male wing size remain not significant, although the general trend that *droncI29* wings are slightly smaller. While there could be sex-specific differences in the capacity to regenerate that contribute to this observation, it is unclear what the underlying mechanism could be.

Reviewer #1 (Recommendations for the authors):

The work in this paper is already very complete and very well worked out. The conclusions are well supported by the data in this manuscript. I do not have any experimental requests, only a few minor and formal requests/questions.

(1) Why does Diap1 overexpression not affect regenerative proliferation, whereas mir(RHG) and droncI29 do, given that Diap1 acts between RHG and Dronc?

We speculate on this point in the discussion section but have adjusted some of the phrasing for clarity.

(2) I assume that the authors used the cleaved Dcp-1 antibody from Cell Signaling Technologies. I recommend that the authors refer to this antibody as cDcp-1 in text and figures as this antibody specifically detects the cleaved, and thus activated form of Dcp-1, and not the uncleaved, inactive form of Dcp-1 which has a uniform expression in the discs.

Changed to cDcp-1.

(3) Line 299: Hay et al. 1994 did not show that p35 inhibits Drice and Dcp-1 (in fact, both genes were not even cloned yet). This was shown by Meier et al. 2000 and Hawkins et al. 2000. Please correct references.

Corrected.

(4) Line 574/575. Meier et al. 2000 did not show that Dronc is mono-ubiquitylated. This was shown by Kamber-Kaya et al., 2017. Please correct.

Corrected.

Reviewer #2 (Recommendations for the authors):

(1) Does domeless knockdown cause apoptosis without tissue ablation (Figures 2C-E)? Currently, the non-ablation control is not shown.

Domeless knockdown does not cause apoptosis in the absence of ablation (Added Figure 2 – Figure supplement 1A).

(2) The supplemental experiment with zfh2-RNAi is hard to interpret because there is no evidence of RNAi knockdown based on the staining with the anti-Zfh2 antibody.

As noted above, a longer *zfh-2* knockdown does not appear to alter Zfh-2 protein levels. A quantification of posterior NiA/NiCP following knockdown shows a slight (non-significant)

increase in posterior NiA/NiCP. Considering these new results, we have altered our interpretation within the appropriate results and discussion sections.

(3) The authors should consider adding a diagram showing where mir(RHG) and DIAP1 are in the apoptotic/caspase activation pathway (Figure 7N).

Completed, Figure 7N and 7O.

Reviewer #3 (Recommendations for the authors):

(1) Figure 2 I -The purported increase in NiA should be quantitated relative to the NiA in G across many discs.

Completed (Figure 2L)

(2) Figure 2 M - contrary to the conclusion drawn, the posterior Dcp1 does not appear different from that in the control (K). This conclusion that the NiA does not occur in the margin could be better supported with more images/quantification.

We have exchanged the image for a representative one that more clearly shows the lack of margin NiA and highlighted with an arrowhead (Figure 2K)

(3) Figure 2 supp 1 E - the "slight increase" in NiA in the pouch is relative to which control? Can this conclusion be supported by quantification?

Figure 2L now quantifies this change.

(4) Figure 2 Supp 1 D, E - these discs supposedly have Zfh2 RNAi expressed, but there appears to be no reduction in Zfh2.

We were unable to demonstrate a reduction of Zfh2, even with a longer knockdown. Considering these new data, we have altered our conclusions from the Zfh2 experiments.

(5) Figure 2 Supp 1 I - please quantitate the Dcp-1 across many discs to support the conclusion.

This is the UAS-wg experiment, which we decided to remove from the quantification given the non-specific increase in cDcp-1 throughout the disc (likely as a result from ectopic Wg expression).

(6) Figure 4 legend M - The authors conclude that the experiment indicates that "NiA promote proliferation independent of AiP". It would be more precise to say that NiA cells do not secrete AiP mitogens and do not increase the proliferation of surrounding cells when prevented from completing apoptosis. To say that the NiA-induced proliferation does not require AiP would require eliminating AiP, perhaps through reaper hid grim knockdown or mitogen knockdown.

Corrected.

Minor concerns and clarification needed:

(7) Line 61 - consider the distinction between a feed-forward loop and a positive feedback loop.

Corrected.

(8) Line 338 - it would be helpful to have a brief explanation of what the GC3Ai consists of and how it reports caspase activity.

Corrected.

(9) Line 343 - the authors should clarify by what they mean when they state GC3Ai-positive cells are "associated with" mitotic cells. Are the GC3Ai cells undergoing mitosis? Or is the increase in mitosis non-autonomous?

Adjusted. "associated with adjacent proliferative cells".

(10) Lines 392-394 - the authors should add brief descriptions of how the Drice-Based sensor and the CasExpress function, so the readers can better understand the distinctions between these sensors and the previously mentioned sensors (anti-Dcp1 and GC3Ai). In addition, please clarify how the Gal80ts modulates the sensitivity of the CasExpress.

Descriptions of DBS and CasExpress and additional clarification provided.

(11) Line 413: How does Gal80ts suppress the background developmental caspase signal, and how does this suppression lead to NiCP cells expressing GFP?

This section has been reworded to clarify.

(12) Line 417 - which GFP label is referred to here?

This section has been reworded to clarify.

(13) Line 445 is the first mention of the CARD domain - it could be introduced more fully and explained why the DroncDN's lack of effect on proliferation excludes the CARD domain as being important.

Clarified. See also the discussion for the significance of the CARD domain as dispensable for regenerative proliferation following necrosis.

(14) Line 452 - "As mentioned" - the manuscript has not previously mentioned DIAP1 modification of the CARD domain and what that modification does. Perhaps the previous explanatory text was inadvertently removed?

Corrected.

(15) The Discussion is a lengthy list of experiments that the authors did not do or observations they were unable to make. This section could benefit from a more in-depth discussion of necrosis and the possibility that NiCP cells contribute to repair after injury across contexts and species.

We have made several changes to the discussion that elaborate on some of the points listed in the public reviews.

(16) All figures: Consider making single-channel panels grayscale to aid visualization. Also consider using color combinations that can be distinguished by color-blind readers.

We appreciate these suggestions and will consider them for future manuscripts.

| (17) All figure legends - are error bars SD or SEM?

Standard deviation. Added to appropriate legends.

| (18) Figure 1A,C - it would be helpful in the diagrams to note when the necrosis occurs/completes.

The endpoint of necrosis is not well defined, given the simultaneous changes that occur with regeneration. Thus, we opted to not include an indicator of when necrotic ablation ends.

| (19) Figure 1B - it would be helpful to name the GAL4 drivers whose expression domain is depicted to correlate with the terms used in the text.

Completed.

| (20) Figure 1 legend- what do the different colors of the arrowheads denote? The dotted lines are in R' and S', not N' and O'.

Completed.

| (21) Figure 2G - the yellow dashed line is not in the same place in the two images.

Corrected.

| (22) Figure 2I - what is the open arrowhead?

Completed (Figure 2I legend).

| (23) Figure 3 legend - please describe what the time course is observing (EdU).

Completed.

| (24) Figure 4 - please include the yellow boxes in the Dcp-1 channels.

Completed.

| (25) Figure 5 F' - add the arrowheads to all the panels. The yellow arrowhead appears to be pointing to nothing.

Completed.

| (27) Figure 5 legend - what is a "cytoplasmic undisturbed cell"? What is the arrowhead in G? J and J' should show the same view at different time points or different views at the same time point.

Figure legend has been corrected.

| (28) Figure 5 Supp 1 would be especially helped by having more single-channel panels in grayscale.

For clarity and consistency, we chose to maintain the different color channels.

(29) Figure 5 Supp 1 D and E - It would be helpful to have higher magnification and arrows pointing to the cells of interest. Why are there TUNEL+ cells that do not have caspase activation (green)?

We have added arrowheads as suggested. We believe the disparity in TUNEL and GC3Ai signals are a result of the different sensitivities of the IF staining and the TUNEL assay.

(30) Figure 5 Supp 1 F - perhaps the arrowheads should be in all panels - they point to empty spaces with no H2Av staining in the final panel. Perhaps a higher magnification image would make the "strong overlap" of the two signals more apparent?

We have added arrowheads where appropriate.

(31) Figure 6 D-E - does the widespread GFP lineage tracing signal suggest that most cells in the repaired tissue originated from cells that once had caspases activity?

Possibly, however given that CasExpress leads to significant developmental labeling, we were unable to determine to what extent the signal in this experiment comes from NiA/NiCP activity versus developmental labeling. Note that tubGAL80ts is not present in this experiment.

(32) Writing corrections:

Line 343 "positive" is misspelled.

Completed

Line 429 - a word may be missing.

Completed

Line 639 - the word "day" may be missing.

Completed

Line 658 - what temperature was the recovery?

Completed

Lines 706-708 - were the discs incubated in 55 mL and 65 mL of liquid, or a smaller volume?

Completed

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